INCH-POUND

MIL-HDBK-117
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MILITARY HANDBOOK

MOTORS, ELECTRIC, SMALL
ACQUISITION, OPERATIONS AND SUPPORT
FOR PRIMER



AMSC N/A

FSC 6105

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20301

PRIMER - Acquisition, Operations and Support for Small Electric Motors

1. This standardization handbook was developed by the Department of Defense with the assistance of the military departments, federal agencies, and industry.

2. This document provides useful information to all persons involved daily or frequently with small electric motors. It does not supplement any departmental directive, regulation, manual, specification, or standard.

3. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U.S. Army Belvoir Research, Development and Engineering Center (STRBE-TSE), Fort Belvoir, VA 22060-5606, by using the Standardization Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. This MIL-HDBK is directed to the attention of all persons who are directly or indirectly concerned with the acquisition and operations and support of small electric motors.

2. Small electric motors are used for a multitude of very important applications in military weapon systems and equipment. Often taken for granted, these mostly unseen motors, because of their size, may be found, for example, in:

- Aircraft flight control systems.
- Ground and airborne radar systems.
- Air traffic control equipment.
- Electronic equipment air conditioners.
- Combat vehicle micro-cooling and collective protection systems.
- Fuel pumps.

3. Field failures of small electric motors have adversely affected the operational readiness of some weapons and early warning defense systems. Mission disruption, mission failure, personnel incapacitation, and ultimately loss of system end items and loss of life are the prices paid for these motor failures. Such losses are unbearable under any condition.

4. Because of their concern for small electric motor field failures, the Department of Defense (DOD) Defense Standardization Program Office (DSPO) commissioned an engineering evaluation of field failures of small electric motors rated at one horsepower or less. This evaluation, started in 1984 and completed in 1985, showed that the problem of field failures appeared to be due to two principal failure modes:

- Premature failures - motor operational life was much less than stated in applicable requirement and technical documents.

- Excessive failures - motor failure rates greatly exceeded those anticipated by equipment developers and users.

Within these two modes, the field failures were attributable to a variety of reasons including problems with:

- Design.
- Technical data.
- Hardware testing.
- Personnel skills.
- Configuration management.
- Contractor selection.
- Contractor performance.
- Defective products.
- Deficiency reporting.
- Quality control.
- Operational use and abuse.
- Maintenance.
- Storage.
- Repair, overhaul rebuild actions.

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- Tools and facilities.
- Former government surplus equipment.

One conclusion of the DSPO engineering evaluation was that no single discipline was at fault - all shared equally in their "contribution" toward generation of field failures. In a sense, this negative contribution resulted in the small motor acquisition process being unable to consistently provide replacement motors and repair parts to users. And, the users were unable to consistently keep their small electric motor assets operational and repairable, thus, idling the equipment in which they were installed.

5. This MIL-HDBK is a direct result of the 1984-1985 DSPO engineering evaluation of small electric motors. It focuses upon available design data, case studies, and lessons learned. A small electric motor decision network is used to take the reader through typical steps in acquiring, operating, and supporting small electric motors. These steps, while not represented in a time-phased system life cycle manner, do raise questions and provide information which may assist personnel in their discipline involved with small electric motor acquisition, operation, and support to reduce field failures of these motors.

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1. SCOPE

1.1 Purpose. The purpose of this handbook is to provide general information for use in reducing field failures of small electric motors.

1.2 Application. Persons whose responsibilities concern the acquisition and operations and support of small electric motors may use this handbook during the performance of their duties.

1.3 Level of information. Information in this handbook in general in content and does not amend or supplement existing guidance or regulatory documents. No attempt has been made to provide a handbook which teaches or instructs persons how to perform their duties.

1.4 Handbook structure. This handbook is structured around references for motor design and specifications, results of an engineering evaluation of field failures of small electric motors to include case histories and lessons learned, and a decision network. The decision network, in turn, raises questions and discusses the process of acquiring, operating, and supporting small electric motors.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS)1 and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

(Copies may be obtained from General Services Administration, Specification and Consumer Information Distribution (WFSIS), Washington Navy Yard, Bldg. 197, Washington, D.C. 20407).

MILITARY

MIL-M-7969	- Motors, Alternating Current, 400-Cycle, 115/200-Volt System, Aircraft, General Specification For.
MIL-M-8609	- Motors, Direct Current, 28-Volt System Aircraft, General Specification For.
MIL-M-13786	- Motors, Fractional Horsepower, Direct Current and Universal. (For Communication and other electronic and Special Military Applications.)
MIL-M-13787	- Motors, 60-Cycle, Alternating Current, Fractional H.P. (Shipboard Use.)
MIL-M-17059	- Motors, 60-Cycle, Alternating Current, Fractional H.P. (Shipboard Use.)
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-M-175561	- Motor, Direct-Current, Fractional H.P. (Shipboard Use.)
MIL-M-22387	- Motor, Alternating Current.
MIL-B-23071	- Blowers, Miniature, For Cooling Electronic Equipment, General Specification For.
MIL-T-60530	- Technical Data Package for AMC Material.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-961	- Military Specifications and Associated Documents, Preparation of.
MIL-STD-962	- Military Standards and Handbooks, Preparation of.
MIL-STD-1388-1	- Logistic Support Analysis.
MIL-STD-1388-2	- DOD Requirements for a Logistic Support Analysis Record.
MIL-STD-1412	- Motors, Electric; For Air Conditioner Applications, Methods of Test and Instructions.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVSEA 56Z32/HJB 9302 - Guide to the Preparation of a Procedure to Obtain Certification for Providing Squirrel-Cage Induction Motors with a Sealed Insulation System.

DARS NO. 6 - Analysis, 19 June 1984.

DARS NO. 6, 1 June 1983 - DOD Replenishment Parts Breakout Program.

DLAR 3200.1, 4 January 1979 - Engineering Support For Items Supplied by Defense Logistics Agency and General Services Administration.

DLAR 4155.24, 30 July 1979 - Reporting of Product Quality Deficiencies Across Component Lines.

DLAR 8260.2, 21 March 1984 - Materiel Deficiency Investigation and Reporting System.

FAR Part 46 - Quality Assurance.

Declared Excess Personal Property Listing.

Electric Motor Reliability Model Report, December 1977 - RADC-TK-77-408.

DGSC Motor Problem Data Base, Revised April 1984.

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Government Accounting Office (GAO) Letter B-214275, 19 June 1984 - "Analysis of Selected Sections of DOD's Revised Breakout Regulation."

(Requests for copies should be addressed to U.S. Army Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FE, Fort Belvoir, VA 22060-5606).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators.

(Copies may be obtained from Orders Department, National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, D.C. 20037.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

ANSI C50.21-2972/IEEE Std 114 - IEEE Test Procedure for Single-Phase Induction Motors.

ANSI/IEEE Std 112 - IEEE Standard Test Procedure for Polyphase Induction Motors and Generators.

(Copies may be obtained from the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

UNDERWRITERS' LABORATORIES (UL)

UL 674/ANSI C33 72 - Electric Motors and Generators For Use in Hazardous Locations, Class 2, Groups C and D.

(Copies may be obtained from the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Manufacturer's Literature File - 17 Items

(Information and documents may be obtained from U.S. Army Belvoir Research, Development and Engineering Center, ATTN: STRBE-FE, Fort Belvoir, VA 22060-5606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Assembly. A number of parts or subassemblies of any combination thereof joined together to perform a specific function and capable of disassembly. (Examples: Power shovel-front, fan assembly, audio frequency amplifier.) NOTE: The distinction between an assembly and subassembly is determined by the individual application. An assembly in one instance may be a subassembly in another where it forms a portion of an assembly. (MIL-STD-1388-2)

3.2 Attaching part. An item used to attach assemblies or parts to the equipment or to each other. (MIL-STD-1388-2)

3.3 Component. An assembly or any combination of parts, subassemblies and assemblies mounted together normally capable of independent operation in a variety of situations. (MIL-STD-1388-2)

3.4 Corrective maintenance. All actions performed as a result of failure to restore an item to a specified condition. Corrective maintenance can include any or all of the following steps: Localization, Isolation, Disassembly, Interchange, Reassembly, Alignment, and Checkout. (MIL-STD-1388-1)

3.5 Defense Acquisition Circular (DAC). A Department of Defense (DOD) circular which amends the Federal Acquisition Regulation (FAR).

3.6 Defense Acquisition Regulation (DAR). The DOD acquisition regulation now superseded by the FAR, DOD FAR Supplement, and DAC's.

3.7 DAR Supplement (DARS). A DOD supplement to the now superseded DAR.

3.8 Data Item Description (DID), DD Form 1664. A form used to define and describe the data required to be furnished by the contractor. Completed forms are provided to contractors in support of and for identification of each data item listed on the CDRL. (MIL-STD-1388-1)

3.9 Design change. An approved engineering change incorporated into the end item which modifies, adds to, deletes, or supersedes parts in the end item. (MIL-STD-1388-2)

3.10 Design control activity. A contractor or government activity having responsibility for the design of a given part and for the preparation and currency of engineering drawings and other technical data for the part. (DARS No. 6)

3.11 Design parameters. Qualitative, quantitative, physical, and functional value characteristics that are inputs to the design process for use in design tradeoffs, risks analysis, and development of a system that is responsive to system requirements. (MIL-STD-1388-1).

3.12 DOD FAR Supplement. The DOD FAR Supplement contains material that implements the FAR, as well as supplementary material that is unique to the Department of Defense. This supplement is not a stand-alone document and must be read in conjunction with the FAR. It contains directions to DOD contractor personnel as to (i) what provisions, clauses, and cost principles are authorized

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for DOD contracts and (ii) what other procedures and actions they must follow in awarding and administering DOD contracts. (FAR)

3.13 End Article/Product. A component, assembly, or subassembly being procured as the top item on the contract. (MIL-STD-1388-2)

3.14 End item. A final combination of end products, component parts, and/or materials which is ready for its intended use, e.g., ship, tank, mobile machine ship, aircraft, receiver, rifle, or recorder. (MIL-STD-1388-2)

3.15 Facilities. The permanent or semipermanent real property assets required to support the material system, including conducting studies to define types of facilities or facility improvements, locations, space needs, environmental requirements, and equipment. One of the principal elements of ILS. (MIL-STD-1388-1)

3.16 Federal Acquisition Regulation (FAR). The federal regulation used to acquire material, goods, and services for federal departments and organizations.

3.17 Government Furnished Material (GFM). Material provided by the Government to a contractor or comparable Government production facility to be incorporated in, attached to, used with or in support of an end item to be delivered to the Government or ordering activity, or which may be consumed or expended in the performance of a contract. It includes, but is not limited to, raw and processed materials, parts, components, assemblies, tools, and supplies. Material categorized as Government Furnished Equipment (GFE) and Government Furnished Aeronautical Equipment (GFAE) are included. (MIL-STD-1388-1)

3.18 Integrated Logistic Support (ILS). A disciplined approach to the activities necessary to (a) cause support considerations to be integrated into system and equipment design, (b) develop support requirements that are consistently related to design and to each other, (c) acquire the required support, and (d) provide the required support during the operational phase at minimum cost. (MIL-STD-1388-1)

3.19 Logistic Support Analysis (LSA). The selective application of scientific and engineering efforts undertaken during the acquisition process, as part of the system engineering and design process, to assist in complying with supportability and other ILS objectives. (MIL-STD-1388-1)

3.20 Logistic support analysis documentation. All data resulting from performance of LSA tasks conducted under this standard pertaining to an acquisition program. (MIL-STD-1388-1)

3.21 Logistic Support Analysis Record (LSAR). That portion of LSA documentation consisting of detailed data pertaining to the identification of logistic support resource requirements of a system/equipment. See MIL-STD-1388-2 for LSAR data element definitions. (MIL-STD-1388-1)

3.22 Maintenance concept. A narrative description identifying the broad, planned approach to be employed in sustaining the system/equipment at a defined level of readiness or in a specified condition in support of the operational requirement. Initially stated by the requiring authority for design and support planning purposes and is expanded by performing activity prepared inputs during

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full-scale development. Provides the basis for the maintenance plan. Usually includes guidelines pertaining to projected maintenance tasks, levels, and locations: organic/contractor maintenance workload mix; condition monitoring; fault isolation and testing approach; compatibility with existing support and test equipment, etc. May be influenced or modified as system/equipment development proceeds. (MIL-STD-1388-2)

3.23 Maintenance levels. The basic levels of maintenance into which all maintenance activity is divided. The scope of maintenance performed within each level must be commensurate with the personnel, equipment, technical data, and facilities provided. (MIL-STD-1388-1)

3.24 Maintenance planning. The process conducted to evolve and establish maintenance concepts and requirements for a materiel system. One of the principal elements of ILS. (MIL-STD-1388-1)

3.25 Manpower. The total demand, expressed in terms of the number of individuals, associated with a system. Manpower is indexed by manpower requirements, which consist of qualified lists of jobs, slots, or billets that are characterized by the descriptions of the required number of individuals who fill the job, slots, or billets. (MIL-STD-1388-1)

3.26 Manpower and personnel. The identification and acquisition of military and civilian personnel with the skills and the grade required to operate and support a materiel system at peacetime and wartime rates. One of the principal elements of ILS. (MIL-STD-1388-1)

3.27 Organizational equipment/unit-organizational maintenance. Maintenance which is the responsibility of and performed by the using organization on its assigned equipment. Its phases normally consist of inspecting, servicing, lubricating, adjusting, and replacing of parts, minor assemblies and subassemblies. (MIL-STD-1388-2).

3.28 Overhaul. That maintenance effort (service/action) prescribed to restore an item to completely serviceable/operational condition as required by maintenance standards in appropriate technical publications. Overhaul is normally the highest degree of maintenance performed. (MIL-STD-1388-2)

3.29 Packaging, handling, storage, and transportation. The resources, processes, procedures, design considerations, and methods to ensure that all system, equipment, and support items are preserved, packaged, handled, and transported properly including: environmental considerations and equipment preservation requirements for short and long term storage and transportability. One of the principal elements of IlLS. (MIL-STD-1388-1).

3.30 Part. One piece, or two or more pieces, joined together which are not normally subject to disassembly without destruction or impairment of designed use. (MIL-STD-1388-2)

3.31 Personnel. The supply of individuals, identified by specialty or classification, skill, skill level, and rate or rank, required to satisfy the manpower demand associated with a system. This supply includes both those individuals who support the system directly (i.e., operate and maintain the system), and those individuals who support the system indirectly by performing

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those functions necessary to produce and maintain the personnel required to support the system directly. Indirect support functions include recruitment, training, retention, and development. (MIL-STD-1388-1)

3.32 Preventive maintenance. All actions performed in an attempt to retain an item in specified condition by providing systematic inspection, detection, and prevention of incipient failures. (MIL-STD-1388-1)

3.33 Prime contractor. A contractor having responsibility for design control and/or delivery of a system/equipment, such as aircraft, engines, ships, tanks, vehicles, guns and missiles, ground communications and electronic systems, ground support, and test equipment. (DARS No. 6)

3.34 Provisioning. The process of determining and acquiring the range and quantity (depth) of spares and repair parts and support and test equipment required to operate and maintain an end item of materiel for an initial period of service. (MIL-STD-1388-1)

3.35 Provisioning Technical Documentation (PTD). That documentation furnished by contractors or prepared by a Department of Defense activity which is used by a Government activity for identification, determination of initial requirements, cataloging, and contractual formalization of items to be procured through the provisioning process. PTD refers principally to provisioning lists, priced spare parts lists, decks of electric accounting machine provision cards, and electronic data processing equipment tapes. Its also includes supplementary technical documentation, such as drawings, sketches, and brief item descriptions.

3.36 Rebuild. Those services/actions necessary to the restoration of unserviceable equipment to a like-new condition in accordance with original manufacturing standards. Rebuild is the highest degree of materiel maintenance applied to equipment. The rebuild operation includes the action of returning to zero those age measurement (hours/miles, etc.) considered in classifying equipment/components. (MIL-STD-1388-2)

3.37 Remove. A task function to perform operations necessary to take a spare/repair out of the next larger assembly. (MIL-STD-1388-2)

3.38 Remove and replace. A task function to substitute a serviceable spare/repair part for a malfunctioned, damaged, or wornout part. (MIL-STD-1388-2)

3.39 Repair. A task function utilized as a corrective maintenance action or task function to restore to a serviceable condition and end item, assembly, subassembly, module, or component. Also to be utilized as maintenance action or task function to restore an item removed from the end item through replacement of lower-order nonrepairable items and through rework such as patching, welding, grinding, straightening, facing, machining, or resurfacing to correct a specific fault. Repair actions will include discrete sequences for fault location, correction of the fault or malfunction, and verification that the fault has been corrected. (MIL-STD-1388-2)

3.40 Repair parts. Those support items that are an integral part of the end item or system which are coded as nonrepairable. (MIL-STD-1388-1)

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3.41 Replace. To substitute serviceable items for malfunctioned, damaged, or wornout items. (MIL-STD-1388-2)

3.42 Replenishment part. A part, repairable or consumable, purchased after provisioning of that part, for replacement, replenishment of stock, or use in the maintenance overhaul, and repair of equipment, such as aircraft, engines, ships, tanks, vehicles, guns and missiles, ground communications and electronic systems, ground support, and test equipment. As used in this supplement, except when distinction is necessary, the term "part" includes subassemblies, components, and subsystems. (DARS No. 6)

3.43 Scheduled maintenance. Preventive maintenance performed at prescribed points in the item's life. (MIL-STD-1388-1)

3.44 Source, Maintenance, and Recoverability (SMR) Codes. Uniform codes assigned to all support items early in the acquisition cycle to convey maintenance and supply instructions to the various logistic support levels and using commands. The uniform code format is composed of three, two character parts, Source Codes, Maintenance Codes, and Recoverability Codes in that order. (MIL-STD-1388-1)

3.45 Spares. Those support item that are an integral part of the end item or system which are coded as repairable. (MIL-STD-1388-1)

3.46 Specification. A document prepared specifically to support acquisition which clearly and accurately describes essential technical requirement for purchased materiel. Procedures necessary to determine that the requirements for the purchased material covered by the specification have been met shall also be included. A military specification covers systems, subsystems, items, materials, or products that are intrinsically military in character or are used in, or in support of, weapons systems and involve an essential system function or interface. (MIL-STD-961)

3.47 Standard. A document that establishes engineering and technical requirements for items, equipment, processes, procedures, practices, and methods that have been adopted as standard. Standards may also establish requirements for selection, application, and design criteria for materiel. (MIL-STD-962).

3.48 Subassembly. Two or more parts which form a portion of an assembly or a component replaceable as a whole, but having a part or parts which are individually replaceable. (Examples: Gun mount, stand, window, recoil mechanism, floating piston, telephone dial, IF strip, mounting board with mounted parts, power shovel, dipper stick). (MIL-STD-1388-2)

3.49 Supply support. All management actions, procedures, and techniques required to determine requirements for, acquire, catalog, receive, store, transfer, issue, and dispose of secondary items. This includes provisioning for initial support as well as replenishment supply support. One of the principal elements of ILS. (MIL-STD-1388-1)

3.50 Supportability. The degree to which system design characteristics and planned logistics resources including manpower meet system peacetime operation and wartime utilization requirements. (MIL-STD-1388-1)

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3.51 Supportability related design factors. Those supportability factors which include only the effects of an item's design. Examples include inherent reliability and maintainability values, testability values, transportability characteristics, etc. (MIL-STD-1388-1)

3.52 Support concept. A complete system level description of a support system, consisting of an integrated set of ILS element concepts, which meets the functional support requirements and is in harmony with the design and operational concepts. (MIL-STD-1388-1)

3.53 Support equipment. All equipment (mobile or fixed) required to support the operation and maintenance of materiel system. This includes associated multiuse end items, ground handling and maintenance equipment, tools, metrology and calibration equipment, communications resources, test equipment, and automatic test equipment with diagnostic software for both on and off equipment maintenance. It includes the acquisition of logistics support for the support and test equipment itself. One of the principal elements of ILS. (MIL-STD-1388-1)

3.54 Support resources. The materiel and personnel elements required to operate and maintain a system to meet readiness and sustainability requirements. New support resources are those which require development. Critical support resources are those which are not new buy require special management attention due to schedule requirements, cost implications, known scarcities, or foreign markets (MIL-STD-1388-1)

3.55 Support system. A composite of all the resources that must be acquired for operating and maintaining a system or equipment throughout its life cycle. (MIL-STD-1388-1)

3.56 System engineering process. A logical sequence of activities and decisions transforming an operational need into a description of system performance parameters and preferred system configuration.

3.57 System/Equipment. The item under analysis, be it a complete system, or any portion thereof being procured. (MIL-STD-1388-1)

3.58 Technical data. Recorded information, regardless of form or characteristic, of a specific or technical nature. It may be, for example, document research, experimental, developmental, or engineering work, or be usable or used to define a design or process or to procure, produce, support, maintain, or operate materiel. The data may be graphic or pictorial delineations in media, such as drawings or photographs; text in specifications or related performance or design type documents; or computer printouts. Examples of technical data include research and engineering data, engineering drawings and associate lists, specifications, standards, process sheets, manuals, technical reports, catalog item identifications and related information, and computer software documentation. Technical data does not include computer software or financial, administrative, cost and pricing, and management data or other information identical to contract administration. (DAC #84-1)

3.59 Technical data package. A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable

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technical data, such as plans, drawings, and associated lists, specifications, standards, models, performance requirements, quality assurance provisions, and packaging data. (MIL-T-60530 (MU))

3.60 Unscheduled maintenance. Corrective maintenance required by item conditions. (MIL-STD-1388-1)

3.61 User. The Major Command designated to receive a system from a Materiel Developer (MATDEV) for accomplishing an assigned operational mission under a Table of Organization and Equipment (TOE), Table of Distribution and Allowances (TDA), or other enabling document.

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4. GENERAL INFORMATION

4.1 Introduction. Field failures of low horsepower electric motors have adversely affected the operational readiness of some weapon and early warning defense systems. Electric motor failures, as a minimum, could lead to mission disruption and then progress to mission failures, crew member incapacitation, and ultimately loss of life and system end items.

Personnel of the Department of Defense (DoD) Defense Standardization Program Office (DSPO) expressed concern for the field failures of small electric motors and commissioned a survey to determine the causes of motor failures. The results of that survey showed that: (1) the acquisition system failed to consistently provide satisfactory motors and repair parts, (2) users abused and improperly maintained motors, (3) all disciplines associated with small electric motors contributed equally concerning the failure problem.

The remainder of this section, paragraphs 4.2 through 4.5 inclusive, is information which was part of the survey report for DSPO.

4.2 Scope of survey. The scope of the survey was limited to motors of not more than one horsepower. Work included:

- Performing survey of pertinent literature.
- Conferring with military and industry personnel to determine problems with electric motors.
- Evaluating information from appropriate DoD and industry installations involved with small electric motors to determine cause and effect relationships of the field failures.
- Formulating the framework of general outline of viable solutions which would be effective for preventing similar deficiencies causing small motor failures.

4.3 Information data base.

4.3.1 Literature and data base survey. Information researched included general technical information concerning motor design, application, maintenance, and repair of small electric motors. The technical data bases of the National Technical Information Service (NTIS), Institution of Electrical Engineers (IEE) Information Service in Physics, Electrotechnology and Control (INSPEC), and Engineering Index were assessed for a time period of 1975 through 1984. Also, the University of Central Florida on-line catalog was assessed and abstracts of pertinent data were made for later analysis and use.

Twenty-five manufacturers of small specialty electric motors were contacted. Requests were made to each for technical literature to include specifications and descriptive literature.

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4.3.2 Conferences and meetings. Conferences and meetings with technical personnel were held at:

Belvoir RD&E Center (BRDEC), Fort Belvoir, VA
Defense Contract Administration Services (DCAS) Residency, Albuquerque, NM
Defense General Supply Center (DGSC), Richmond, VA
Sacramento Air Logistics Center (SM-ALC), McClellan AFB, CA
Warner Robbins Air Logistics Center (WR-ALC), Robbins AFB, GA
VSE Corporation, Alexandria, VA
Westinghouse, Baltimore, MD
Wilco Electronics, Hackensack, NJ

4.3.3 Information evaluation and retention. Information evaluated, written and oral, was acquired from the actions discussed in paragraphs 4.3.1 and 4.3.2. The acquired information constituted the data base for the survey. It consisted of: 4 commercial motor specifications, 12 Federal/Military/Military organization specifications and standards, 7 Federal Government/DoD procurement and quality assurance regulations, 1 electric motor reliability model technical report, 17 technical packages of motor manufacturer literature, meeting and conference notes, 1 motor problem data base computer run, and 9 miscellaneous documents and message files related to motor failure problems.

4.4 Discussion of data and case histories.

4.4.1 Commercial specifications.

NEMA MG 1-1978 "Motors and Generators" is the most comprehensive single document available to the motor specification engineer to assure adequate design. This specification is supplemented by UL 674, safety standard for motors used in hazardous locations. UL 674 is concerned primarily with the mechanical configuration of the motor, its lubrication, and material treatments.

Standard commercial NEMA motors provide service life measured in years according to information supplied by WR-ALC and previously referenced publications. SM-ALC also reports few problems with NEMA standard motors, indicating that the greater the deviation from the NEMA standards, the greater the likelihood of problems. They did, however, point out a single case in which a NEMA replacement motor did not fit the next higher assembly because of finds on the motor housing.

Two other important commercial documents are ANSI Standard C50.21-1972 (IEE Std 114-1969, "IEE Test Procedure for Single Phase Induction Motors"), and ANSI Standard 112-1978 (IEE Std 112-1978 "IEE Standard Test Procedure for Polyphase Induction Motors and Generators"). These test procedures should be considered when establishing component equivalency programs for the defining of motors equal to qualified NSN's for supply replenishment.

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4.4.2 Federal, military, military organization specifications and standards.

Federal specification CC-M-1807 for fractional and integral horsepower AC motors is primarily an abstract of NEMA MG 1, with some workmanship, environmental, and special government material and marking requirements added. It provides some guidelines on motor selection and ordering data. The primary value of this specification is for the ordering of essential standard NEMA motors. It should be used whenever possible.

There are a number of Army, Navy, and Air Force specifications available, most of which were developed 20 or more years ago. For this reason they do not reflect the latest developments in motor technology. Nevertheless, as a group they contain enough, well conceived ideas and constraints to service as the basis for adequate motor procurement in almost any category, provided abstracting and combining of the various specification paragraphs is performed with the specific motor application in mind.

MIL-M-7969 is a performance oriented specification for 400 Hz 115/200-Volt aircraft motors. It provides detailed standardized test procedures, but does not provide sufficient information to permit easy parts breakout for resupply.

The workmanship requirements are not sufficiently detailed to serve as a basis for quality analysis, nor is there guidance on motor specific manufacturing problem areas for quality assurance personnel.

MIL-M-8609 is a general specification for 28-Volt DC aircraft motors. The annotation applied to MIL-M-7969 applies equally to this specification.

The primary value of these two documents to the present program lies in the establishing of performance criteria for airborne motors. The test procedures called out should be considered as a basis for developing mandatory component equivalency tests for resupply motors.

MIL-M-13786 is a performance and test specification for fractional horsepower DC and universal motors. It is approximately in the same category as the previous two specifications with regard to breakout capability and inspection guidance.

Most of the specification requirements are well conceived and should be considered for guidance purposes during the remainder of this program. Again, however, the requirements of the specification are not sufficient to permit parts breakout, nor is there adequate guidance for field inspection personnel.

MIL-M-13787 is similar to MIL-M-13786 except that it applies to fractional horsepower AC squirrel cage motors. These specifications are in part NEMA based with regard to dimensions and materials.

This specification, like those previously discussed, is necessary but not sufficient to assure quality motors entering the supply system. It should be considered in this work primarily as an indication of the development of thought processes concerning motor procurement during the past 30 years.

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MIL-M-17059 covers 60 Hz fractional horsepower motors for shipboard use. This specification is an excellent document, detailing completely performance, test, inspection and quality requirements. It provides for the submission of master drawings and spells out sufficient detail in the master package to permit parts breakout to another competent manufacturer. It also provides detailed instructions for the field inspection to be implemented at all stages of manufacture.

This specification should be considered as a format model for a good procurement document under the existing supply system. The validity and necessity of all of its requirement should be assessed, since it may impose unnecessary requirements in some cases. The only obvious problem with a document of this type is the possibility of its significantly increasing the initial procurement cost of a motor. Discretion in its full application should be observed.

MIL-M-17060 applies to 60 Hz integral fractional horsepower motors. Its content is similar to MIL-M-17059. It is represented here as a reference because of its recent revisions and more recent amendments.

MIL-M-17059 should be used as an adjunct to MIL-M-17060 for upgrading purposes in developing documentation for the present program.

MIL-M-175561 has the same characteristics as MIL-M-17059 except that it applies to DC fractional horsepower motors. It does not, however, provide the detailed in-process inspection instructions of the previous specification.

MIL-M-22387 is limited to special purpose motors of the hysteresis synchronous type used primarily in such mechanisms as tape drives and other constant speed low load applications. It is presented here as an example of a special purpose motor specification for information purposes.

MIL-B-23071 is also presented for information purposes only. It is representative of the specification type used where a motor is an integral part of an assembly.

MIL-STD-1412(ME) defines a series of test methods for evaluating electric motors used for military air conditioner applications. Its importance to this program lies in the concept of using component equivalency testing to evaluate resupply motors offered as equivalent to motors qualified with original systems.

The lack of an overall component equivalency program is partially responsible for the presence in the supply pipeline of unsatisfactory motors. At present a motor offered to DGSC as an "equal" is sent to the system manufacturer for confirmation of equality. The system manufacturer has no clear guidelines as to what constitutes "equal" and, thus, is forced to make a subjective judgement as to what testing is appropriate. The possibility exists for inadequate equivalency testing, permitting introduction and use of an inferior product. The possibility also exists for overtesting, excluding satisfactory alternate suppliers.

SM-ALC reports an equivalency tests on an airborne radar sweep motor requiring "instantaneous" motor reversal upon plugging. A number of "equivalent" motors failed this test. A motor is now under test that meets this requirement, but in

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doing so causes other components of the radar unit to fail mechanically. It is apparent that the original motor (qualified with the system) would not meet the equivalency test.

NAVSEA Document 56Z32/HJB9302 represents initial work by the Navy in the area of component equivalency testing and inspection.

Together with previously referenced documents, it serves as a basis for developing a generalized component equivalency policy for small motors.

4.4.3 Procurement and quality assurance regulations.

The procurement and quality assurance functions within DoD are governed by numerous regulations, supplements, directives, instructions, and letters. The Federal Acquisition Regulation with DoD FAR Supplements provides the foundation for procurement and quality assurance for the DoD acquisition system. These and a very selected few other documents are covered herein as the scope of this evaluation and report does not provide for an extensive review of all available regulations.

Defense Acquisition Regulation Supplement (DARS) No. 6, 1 June 1983 establishes the DoD Replenishment Parts Breakout Program and provides priorities and procedures for use by military departments and defense agencies. Its aim is to improve the acquisition of parts through competition and direct purchase from the part manufacturer.

Joint Regulation - DLAR 3200.1, AR 715-13, NAVMATINST 4120.94C, AFR 400-40, MCO 4000.18B 1(4 January 1979) addresses engineering support for items supplied by DLA and GSA. This regulation requires military departments to provide DLA/GSA with specifications, standards, and other engineering and procurement data required to effectively and economically procure items for military department use.

Declared Excess Personal Property Listing (EPPL) issued periodically by the DLA Defense Property Disposal Service provides notice to subscribers of surplus military property available for reuse, sale, or grant to authorized recipients.

Other procurement and quality assurance regulations analyzed, but not briefed in this section are: US GAO letter, DARS No. 6 Analysis, 19 June 1984; DLAR 8260.2 "Materiel Deficiency Investigation and Reporting System," 21 March 1984; and FAR Part 46 "Quality Assurance."

4.4.4 Electric motor reliability model. This report (107 pages) prepared for Rome Air Development Center, number RADCTK-77-408, December 1977, summarizes the results of a study directed towards the development of a failure rate prediction model for fractional horsepower electric motors. A data bank of over 2000 failures of electric motors was updated and supplemented from a survey of electric motor manufacturers. A regression analysis was conducted on all failures, and two dominating, competing, failure modes emerged. These modes are bearing failures and winding failures.

4.4.5 Manufacturers' literature. Such literature shows individual firms product lines, specifications, technical data to include bearing type and lubricants, materials, operating ranges, etc. Each manufacturer's technical data

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portrays that manufacturer to have the skills and facilities necessary to provide OEM and replenishment motors of adequate quality.

4.4.6 Motor problem data base. Information provided by DGSC lists motor problems for the past 30 months. This information is obtained from the DGSC quality evaluation system data base. The primary purpose of this document is to provide other examples of problem motors. It can also be analyzed for problems concerning: Carriers, Condition, Damage, Documentation, Item Technical Markings, Misdirected Shipments, Overages, Packaging, Shortages, Shortage Standards, Wrong Items, and other discrepancies.

4.4.7 Miscellaneous documents. These documents range from subject of Motor Theory and Practice, Repair and Maintenance, through the Defense In-Plant Quality Assurance Program.

4.4.8 Acquisition system overview. A general understanding of acquisition procedures is needed to identify problem causes and problem solutions.

4.4.8.1 System management. Special motors (which are the primary source of motor problems) are usually described by the system prime contractor in a purchase description (PD) or similar document. This technical data package (TDP) is ordinarily prepared by a specification engineer who is not necessarily knowledgeable in motor fabrication technology. A parallel effort by a system specification engineer for another system in the same facility requiring the same motor performance might result in a totally different TDP.

The TDP usually is not sufficient in itself to guarantee a properly performing motor. Consequently, reliance is placed on the integrity and skill of the original motor manufacturer to "read between the lines" and provide a motor suitable to the application.

4.4.8.2 Spares provisioning and replenishing. It is DoD policy to obtain spare parts whenever feasible by competitive bidding. This policy and its implementation is detailed in Defense Acquisition Regulation Supplement (DARS) No. 6 dated 1 June 1983 "DoD Replenishment Parts Breakout Program". Its implementation is supported by the Competition Advocate's Office at DoD (Office of Undersecretary of Defense for Research and Engineering) and subsidiary organizations such as the Competition Advocacy Directorate at the Air Logistic Centers.

This policy creates some general procurement problems which are outlined in General Accounting Office Letter B214275. It also creates some problems specific to motor procurement which will be discussed later. A questionable feature of this problem is the perceived authority of the Competition Advocate to override engineering judgement with regard to the capability of procuring satisfactory replacement motors from new sources.

It should also be noted that the Small Business Administration (SBA) has statutory authority to pass on the competency of small businesses in most instance. At times the exercise of this authority provides for the overriding of an engineering assessment of competency in some circumstances. These situations can be avoided in many cases by identifying motors as critical items with a TDP not adequate to insure satisfactory open procurement.

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4.4.8.3 Critical application items. Technical characteristics of items managed for and procured by the Defense Logistics Agency (DLA) and General Services Administration (GSA) for the Army, Navy, and Air Force may include the item designation "critical application item." As applicable, this designation is determined by the military activity which prepares the item technical data package (TDP) and provides engineering support to DLA and GSA procurement activities.

The definition of "critical application item" is shown in Joint Regulation DLAR 3200.1, AR 715-13, NAVMATINST 4120.94C, AFR 400-40, MCO 4000.188, of 4 January 1979. As presented in the joint regulation, the definition includes items whose failure would be life-threatening or would adversely affect the accomplishment of a military mission.

A critical application item designation is assigned to an item by appropriate notation on various forms prepared by various engineering activities according to various established procedures. Use of the critical application item designation insures source inspection and an approved manufacturer's quality control system as specified in Federal Acquisition Regulation (FAR) Part 46.203. In addition it assures tighter inspection in accordance with the requirements of MIL-STD-105D. As an alternative, the identification of the motor as "Military-Federal Complex Critical Peculiar," shown in FAR 46.204, table 46-1, will assure better than average inspection.

It will be seen that tight inspection requirements are a necessary, but not sufficient condition to assure adequate motor quality in the case of complex special motors.

4.4.8.4 Parts breakout - provisioning. The decision to break out a motor for procurement, for initial system support stocks, at a competitive level beyond the original motor manufacturer involves a calculated risk. It has been noted that the system contractor TDP is usually not sufficient to guarantee a satisfactory product. It is frequently necessary to rework production drawings, or to create new drawings by reverse engineering.

It is possible to transmit the science of motor manufacturing from one manufacturer to another by this method, but the art and the skills of the personnel are specific to each facility. Thus complete drawings are no assurance of adequate production unless the new manufacturer has facilities, in-house engineering staff, and production personnel experienced in the fabrication of motors similar in complexity and quality.

Also, the original motors are most often qualified with the system. Establishing a motor test procedure adequate to assure equivalent performance involves a number of new judgments. Understating test requirements permits the introduction of an inferior product. Overstating requirements escalates costs and eliminates potential replacement items.

4.4.8.5 Parts breakout - replenishment. Parts replenishment is subject to the same problems as parts provisioning with two notable additions: problems introduced during repair and overhaul; and, problems introduced as a result of the repurchase of former government surplus material previously disposed of by the DLA Defense Property Disposal Service, Battle Creek, MI 49017 (see 4.4.8.5.2).

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4.4.8.5.1 Repairable items. A large number of motors are considered repairable based primarily on initial cost and cost to repair. The system maintenance concept determines the level of maintenance, and the extent of maintenance depends on the complexity of the motor, and the personnel and facility capabilities at a given installation. Under the best conditions a depot may have facilities and personnel equal to those of the original manufacturer. This is not the usual case.

Problem motors can be introduced in significant numbers of maintenance facilities without appropriate equipment, personnel, or quality constraints. These defects may not find their way into the quality data bases because of unsatisfactory adherence to an established defect reporting system. Thus, it can be expected that the number of motors requiring additional maintenance after overhaul is larger than would be indicated by a data base scan.

4.4.8.5.2 Surplus procurement. A large number of items and repair parts are declared surplus and disposed of through DoD Instruction (DODI) 4160.21. These are items and repair parts that have been inactive, with respect to supply requisitions, for a period of time varying with the judgement of system and item managers. Many of these items are purchased and stored by commercial organizations for long periods of time.

Since a number of DoD systems including ships and aircraft have been in service for periods in excess of 20 years, some of the original manufacturers have disappeared or changed their manufacturing orientation. Consequently, a procurement request for parts to support these systems is frequently responded to primarily with offers from industry of items previously declared surplus.

Problem motors or motor parts are introduced into the system at this point by the purchase of obsolete items, repair parts that do not fit the latest item modifications, and the receipt of counterfeit merchandise identified with the original manufacturer's part number.

4.4.8.6 Inspection. Assuming that all decisions through procurement award have been optimum, the acquisition of satisfactory motors still depends on the capability and desire of the manufacturer to produce an adequate product, and on the ability of the DCAS quality assurance representative to evaluate the product adequately at each stage of production.

Because most DCAS quality assurance specialists are responsible for inspection in a number of technological areas, it is highly unlikely that any individual responsible for motor facility and product inspection would have the expertise required to reveal a marginal product from an ethical manufacturer, or a defective product from an unethical manufacturer.

Thus, problem motors are introduced into the system at the inspection stage of production in spite of good administrative controls, including those for quality control. Inadequate inspection instruction and special knowledge, together with slow feedback of field failure problems can permit a large amount of defective products to enter the inventory at this point.

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4.4.9 Problem cases.

4.4.9.1 General. A motor that is designed properly, specified properly, procured properly, manufactured properly, used properly, maintained properly, repaired properly, and managed properly will not cause chronic problems in the supply system.

A lapse in any of these categories will cause a problem. Several data bases containing numerous examples of deficiencies in one or more of these areas were reviewed. Primary emphasis has been placed on the DGSC "Quality Evaluation System" data base since it is most appropriate and more than adequate in developing case histories.

Some of the case histories discussed here are not represented in the printout of that data base either because they do not relate to DGSC managed motors, or because they occurred prior to the period for which the data was accessed (3/82 to 12/84).

4.4.9.2 DGSC quality evaluation system data base. A printout of FSC 6105 (motor) quality problems for the above period revealed 2696 reports of purported trouble. During this period there were approximately 13,500 FSC 6105 items being managed by DGSC.

Examination of the discrepancy codes indicates that problem areas cover the entire spectrum of problems described above. However, the codes are not absolute indicators because the problems are substantially interrelated. For example, an S2 code (quantity received less than requested) might result from a U1 (quality deficiency-contractual noncompliance) problem which in turn might result from a U2 (design deficiency) problem.

4.4.9.3 Case histories. It will be seen from the following case histories that the problems frequently have multiple causes, and the assignment of legal responsibility to a single discipline is not always possible. All case histories have been sanitized to avoid finger pointing and because some histories were not validated by the authors.

4.4.9.3.1 Air conditioning fan motor. Company A placed a contract with Company B to manufacture special air conditioners originally built and qualified for the Air Force by Company C. A high world-wide failure rate was experienced for the evaporator/condenser fan motor.

The original C design (which formed part of the Air Force specification) included an Army source control drawing specifying two approved sources. At the time of the A procurement, the only remaining approved source of the fan motor was Company D.

B supplied these air conditioners with a Company E motor meeting form and fit requirements, but (as time would prove) not function. The E motor was apparently qualified with the system, but began to exhibit bearing failures under desert operating conditions. It was ultimately determined that the major problem cause was probably packing of the fan bearings with improper lubricant.

The problem was compounded when replacement motors under NSN 6105-00-949-8629 were supplied by Company F rather than D. The F motors failed almost immediately

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and had an improper shaft configuration. As stated above, the F Company was not shown as an approved motor source on the source control drawing. To date it is not clear how the F motor was approved for procurement as an NSN 6105-00-949-8629 replacement motor.

The problem was resolved by replacing all installed motors with D motors, but it is not certain that the supply system itself has been purged of defective motors. The DGSC data base reveals 33 failure references to this stock number at present, and these should be obtained for further examination.

It would appear that this situation resulted from a sub-contractor providing an unauthorized substitute motor, a system contractor failing to detect the substitution, and another questionable substitute motor being permitted into the resupply pipeline through different but undefined channels.

4.4.9.3.2 Trim tab actuator motor. These motors were obtained from resupply channels and did not mate properly with the actuator. Removal of a portion of the mounting base was required to get a usable fit. They had been procured to Air Force drawings and (according to the DCAS inspector) met the requirements of those drawings during first article inspection. DGSC therefore considered this to be customer (Air Force) error.

The Air Force took issue with this position since the motors had previously been procured satisfactorily using the same drawings. At this time the drawings are being reviewed to see if they are in fact adequate to insure a suitable motor. The motor in question carries NSN 6105-00-959-8645, and shows three entries in the DGSC data base.

It would appear superficially that this problem was insufficient configuration management (drawing control) of the product, inadequate field inspection, or inadequate manufacturer and government quality control.

There is conjecture that the manufacturer (of both the actuator and motor) made a change to the internal mating surfaces of the combination which did not affect form, fit, and function of the assembly. Assuming an Engineering Change Proposal was processed, the change may have been approved routinely by the actuator item manager (SA-ALC), but the motor item manager at SM-ALC was not notified. Since the motor end bell would now have a new configuration, old motors would no longer fit new actuators, and vice versa. Thus, separate spares would be required to fit old and new actuator assemblies although the NSN would not have changed.

The problem then would be a result of a lack of common understanding as to what constitutes identical form, fit, and function. The difficulty results from the complexity of the procurement system and the lapse of interaction between all groups involved before a far reaching decision is made.

4.4.9.3.3 DC high torque motor. Small DC motors ordered as a manufacturer's part number under NSN 6105-00-131-5224 failed to function properly when installed in the next higher assembly. The motors either hung up or functioned for very short periods of time before failure.

A DGSC review indicated the same type of complaint had been filed previously on several contracts for the item supplied by the same manufacturer. The DGSC data base contains seven failure references to this NSN, at least three of which apply

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to the manufacturer in question. A review is currently being made to determine whether or not this manufacturer's part number should be deleted as suitable for this NSN.

This appears to be a case where either the manufacturer's design or quality is not up to an acceptable standard.

4.4.9.3.4 Pitch trim control motor. The history of the pitch trim control motors for the F-4 fleet defines some new problem areas.

This item (NSN 6105-00-791-4363) was originally qualified with the F4 aircraft. The original aircraft mission profile indicated an appropriate duty cycle of 3-seconds on, 17-seconds off.

For the past 10 years these motors have been failing rapidly (probably less than 200 hours) because of changes in operation from the original mission concept. This device is now used to maintain the aircraft in level flight as fuel is burned, munitions fired, or turbulence encountered. The device is also used occasionally to maintain level flight rather than conventional elevator control.

As a result, the actual duty cycle is considerably higher than called for in the original design concept. The result has been a requirement for two or three replacement motors per year per aircraft for a fleet of several thousand aircraft at a cost of approximately $1000.00 per motor.

These motors were rebuilt, usually by replacement of the stators, because of burnt windings. Attempts to replace stators as spare parts from the original drawings produced bad stators. An Air Force reverse engineered drawing changes the duty cycle to 5-seconds on, 15-seconds off. A contract to the reverse engineered drawings was let to a low bidding manufacturer whose past performance history was reported to be less than desirable. This manufacturer failed for an extended period of time to meet first article test requirements.

As a result of this failure, the diminished supply of spares threatened to ground the aircraft. This situation was finally salvaged by the development of an in-house depot capability to rewind the small complex stator.

The problems here are characterized as a failure to anticipate increasing mission severity with the passage of time; failure by the system contractor to supply detailed working drawings sufficiently adequate to assure valid second source procurement; the difficulty of reverse engineering a product to the extent that an inexperienced contractor could build it; and finally the award of the contract to a contractor whose expertise was not sufficient to compensate for these prior lapses.

4.4.9.3.5 Instrument console motor. Motors of the type used to drive tape mechanisms in instrument consoles were procured by DGSC as spares. The motors were built to a specification control drawing package. The contract was again awarded to a low bidder with an undesirable performance history.

Unusable motors entered the procurement system and generated defect reports. Seven failure documents regarding this NSN (6105-00-997-9098) are in the DGSC data base. Attempts are being made to find some use for the nonconforming

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motors. Future problems with this NSN should be alleviated by the current change of the item drawing from specification control to source control.

The problem here is the admission into the supply system of an inadequate product from a marginal supplier.

4.4.9.3.6 Multiple application motors. The C-141 aircraft uses motor NSN 6105-00-782-0720 in two applications. One is on the engine pylon where it is exposed to relatively moderate environmental conditions. The other application is for fuel shutoff where the motor is located in a non-ventilated area close to the engine on the hot side of the compressor.

The pylon motor presents no problems, exhibiting a life of 10-15 years with about six rebuild operations. The engine motors is rebuildable twice on the average because of heat damage. This problem is being resolved by the development of an upgraded motor for the hot location, with all remaining motors being used as pylon spares.

The problem here resulted from the commendable effort to develop multiple applications for the same motor. It is in the best interest of all concerned to minimize the number of different motors in a system, but care must be taken to observe environmental parameters in different functions for the same item.

4.4.9.3.7 Out-of-warranty motors. Storage of motors in the supply systems presents some warranty problems. An AC motor NSN 6105-00-928-7210 was reported to be unsatisfactory in a number of respects when warehouse-supplied after a 1 year guarantee had expired.

The manufacturer (who may be marginal since he is involved in several other, of the cited examples) stated that the problem was primarily bearing lubrication, that the grease was only guaranteed for 1 year, and may have dried out. This is contrary to the shelf life findings of a Redstone Arsenal investigation entitled "Storage Reliability Analysis Electro Mechanical Devices." Nevertheless, it eliminates manufacturer responsibility for all practice purposes.

In another case, a number of servo control motors NSN 6105-00-553-0796 were found to be unusable due to dirty bearings and improper assembly. The manufacturer took the position that these motors were out of warranty because of long storage, and in any event must have been tampered with since they did not meet his quality standards.

Without reflecting on the merit of the manufacturer's position, it is obvious that a major problem area is the elimination of warranty on stored motors whether or not they were satisfactory on entering the supply system. This is a particular problem with the product of marginal builders and places a substantial burden on preaward surveys and DCAS inspections.

4.4.9.3.8 Maintenance. A substantial portion of the life of repairable motors is determined by the adequacy of maintenance. The primary consideration here is the service of the item at all maintenance levels. Undocumented reports were made of motors being made nonrepairable by cutting off leads during removal from the next higher assembly (NHA), and of motors being improperly overhauled in depots due to inexperienced personnel.

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Due to the nature of the reporting system, the extent of these problems would require analysis of the maintenance data base for the Air Force (MODAS) located at Wright-Patterson AFB.

4.4.9.3.9 Surplus procurement. Undocumented reports indicate that former government surplus material is frequently repurchased by the Government, particularly in support of older systems. Usually the original manufacturer has upgraded his tooling and techniques to the extent he can no longer provide replacement motors to old specifications without undergoing expensive retooling programs.

Under this activity, the Government has received the wrong revision of an item, manufacturing rejects stamped with the appropriate part number, and counterfeit parts stamped with the original manufacturer's part number.

This problem is currently under investigation. Recent surplus procurements specify the surplus must have been made by the original manufacturer rather than merely bearing that part number. A more careful evaluation of surplus motors is now being carried out before purchase.

While the volume of motors involved here may not be relatively large; some quantity of "junk" motors are entering the supply chain through this channel.

4.4.9.3.10 Abuse problems. Motors in some application are destroyed due to abuse by untrained or careless personnel. An example of this is the motor used in the T-38 trainer landing light. When these lights are extended or retracted at excessive speeds, the motors overheat to the point of destruction.

While this is not a high volume problem, it is typical of what the system designer might anticipate. In this case, a motor capable of existing for a reasonable period of time with a locked rotor would survive at least until circuit breaker protection could take effect.

4.4.9.3.11 Inspection problems. The elimination of first article testing, the elimination of source inspection in certain cases, insufficient inspection and testing, the lack of specialized training for DCAS quality assurance personnel, and the lack of prompt feedback of field motor problems to inspection personnel all contribute to the introduction of marginal motors into the supply system.

These factors probably played a part in most of the foregoing examples, so that no specific cases will be identified here. Nevertheless, the appropriate indoctrination of DCAS personnel into motor manufacturing technology, and the full use of all contractual quality assurance resources will be required for any major gain in the battle against problem motors.

4.5 Survey conclusions. Conclusions were:

- There are numerous documents pertaining to the design and selection of small electric motors rated at one-horsepower or less.
- Military specifications reviewed are up to 30 years old and tend to rely more upon manufacturer ethics and less on the specifics of detailed manufacturing and quality assurance requirements.

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- Commercial and federal specifications are more current in terms of technology and manufacturing and quality assurance requirements.

- The acquisition process fails to consistently provide satisfactory motors and repair capabilities, despite the existence of numerous and useful guidance documents and regulations.

- All disciplines associated with small electric motors share equally in the acquisition process failure. These disciplines include designers, specification writers, procurement personnel, manufacturers, quality assurance personnel, supply system personnel, users, and repair and disposal personnel.

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5. SMALL ELECTRIC MOTOR DECISION NETWORK

5.1 Purpose of network. Shown in pullout page 83, the small electric motor decision network provides readers with a basis for discussion of problems and information concerning the acquisition and operation and support for small electric motors.

5.2 Network description. The small electric motor decision network consists of 73 numbered blocks. With the exception of block 1, each block either raises a question or commands the completion of a required action. Block 1 identifies the requirement for a small electric motor as part of a new system design.

In appearance, the network may seem to be solely the acquisition life cycle for small electric motors. This is not the intent of the network. While it does address some aspects of acquisition, the network also goes into operations and support activities which follow the motor acquisition process. And, the network is not phased with respect to activity and time. It only represents important activities related to the field failure problems and case histories discussed in section 4, inclusive.

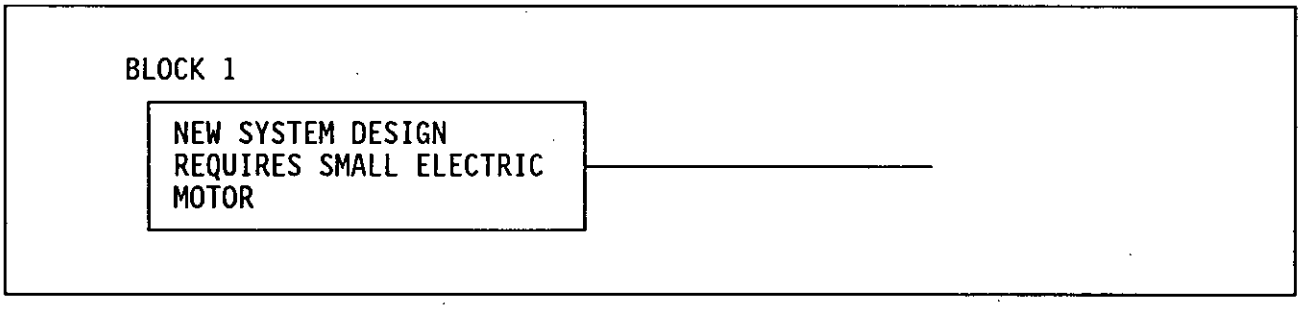
5.3 Network narratives. Each network block has an associated narrative paragraph which provides information and discusses problems peculiar to that block.

In content, each narrative consists of: the purpose of each block; participants, both primary and secondary, involved in block activities; responsibilities of participants; a discussion of problems and applicable solution information; and, a general reference to contract requirements, specifications, standards, government directives and regulations, and manufacturer guidelines and regulations.

By using the information presented in the network narratives, the reader should be able to recognize an opportunity to improve the acquisition and operation and support aspects for small electric motors, thus eliminating problem areas before they occur.

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5.3.1 Block 1. System design.



BLOCK 1

PURPOSE: The purpose of block 1 is to direct the reader's attention to the need for a small electric motor as a component of a contract end article or product.

SECONDARY PARTICIPANT(S): Developmental Engineer (User), System Design Team (Mfr), Logisticians (Mfr, User).

RESPONSIBILITIES: The manufacturer's System Design Engineer is responsible for directing the initiation and completion of a satisfactory system design which meets all user contractual requirements. As the user representative, the Developmental Engineer is responsible for serving as the interface for communicating user requirements to the manufacturer System Design Engineer and System Design Team. The System Design Team is responsible for translating user requirements into a satisfactory system design. Manufacturer Logisticians are responsible for establishing and maintaining a logistics-design interface and Logists Support Analysis (LSA). User Logisticians are responsible for being cognizant of the manufacturer's LSA program.

DISCUSSION: During the conduct of any formal project or program many events must be scheduled and activities completed before that project or program is ended. For purposes of this discussion, events, such as block 2, are milestones which require that decisions must be made or actions have to be taken. Activity, shown as black lines, is the work to be completed or action to be taken before the next event can be undertaken.

In keeping with the aforementioned comments, an assumption is made so that the reader can appreciate the use of the statement in block 1 as the beginning of the small electric motor decision network. The assumption is that the following representative new system events have already happened:

- New system requirement identified.
- Formal new system project established.
- Project funds made available.
- New system contract awarded.
- Government contract surveillance delegated to appropriate government organization.
- Developmental Engineer communicated user's requirements to System Design Engineer, System Design team, and Logisticians.
- Manufacturer's new system design effort initiated.

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- Need for small electric motor recognized.
- New system maintenance concept established.
- Logistic Support Analysis tasks initiated.

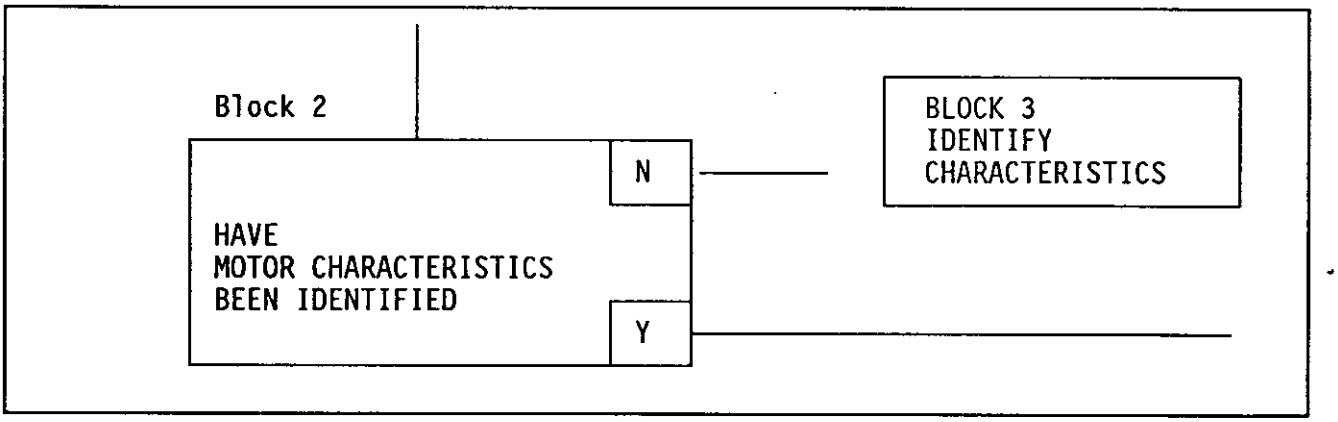
As soon as the need for the small electric motor was established, the manufacturer's System Design Engineer became concerned with determining the motor source of supply. Should the motor be manufactured in-house or under subcontract? At this time, another assumption is made for the reader, i.e., the System Design Engineer, assisted by secondary participants, concluded that the small electric motor should be obtained from a subcontractor.

With this conclusion to obtain the motor from a subcontractor in mind, the reader can now go through the remainder of the small electric motor decision network.

REFERENCES: Contract requirements, specifications, standards, directives, and regulations.

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5.3.2 Block 2 and Block 3. Motor characteristics.



BLOCK 2

PURPOSE: The purpose of block 2 is to show the wide range of electric motor characteristics (service conditions) to be considered by motor designers and motor users.

PRIMARY PARTICIPANT: Motor Designer.

SECONDARY PARTICIPANT(S): System Design Engineer (Mfr), System Design Team (Mfr), Developmental Engineer (User), Logisticians, Motor Manufacturers.

RESPONSIBILITIES: The Motor Designer is responsible for developing the characteristics (service conditions) of the motor to be used in the new system. The System Design Engineer, System Design Team, Developmental Engineer, Logisticians, and Motor Manufacturers are responsible for providing information to the Motor Designer.

DISCUSSION: The success or failure of the new system's small electric motor will depend heavily upon the accurate identification of its required characteristics (service conditions).

Motor designers are faced with an abundance of design data in the form of commercial, federal, and military standards, specifications, handbooks, manufacturer's literature, and other documentation. Some of the more important documents, for purposes of this discussion, are NEMA MG1-1978, CC-M-1807A, and MIL-M-17059A (SHIPS). These documents, and others discussed in paragraphs 4.4.1 and 4.4.2 contain very useful design information for the Motor Designer and others concerned with the selection of electric motors.

Small electric motors should be properly selected with respect to their desired characteristics, both usual or unusual. Usual characteristics relate to: ambient temperature range of 0°C to 40°C; altitude not exceeding 3300 feet (1,000 meters); location and enclosure that give adequate motor ventilation. Unusual characteristics relate to: environment; departure from rated voltage, frequency, or both; speeds other than as rated; poor ventilation; loads such as torsional impacts, overloads, reversing, and frequency of starting. The following list of characteristics, while

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not all inclusive, gives the reader an appreciation of items to be considered when designing a small electric motor:

- Ambient and operating temperatures.
- Voltage, voltage limitations.
- Frequency, frequency limitations.
- Duty cycle - continuous, intermittent, varying, short-time.
- Enclosure - open, waterproof.
- Horsepower, speed ratings, speed variations.
- Torque and current limitations.
- Electrical and mechanical balance.
- Frame and feet.
- Shafts, bearings, rotor and starter core, bearings and lubricants.
- Stresses - shock, vibration, acceleration, axial and lateral loads.
- Access for maintenance.
- Airborne contaminants - dust, salt spray, fog.
- Construction - materials, connections, paint, insulation.
- Life, interchangeability of parts.
- Type and identification plates.
- Mounting position - horizontal, vertical, inclined.
- Direction of rotation.
- End item maintenance concept.

By carefully identifying sufficient and accurate motor characteristics, the Motor Designer can prepare a small electric motor specification for downstream acquisition purposes. This specification should include recommended or approved manufacturing sources, drawings, test requirements, referenced documents, and more. The more detailed this information, the better the selected motor will be.

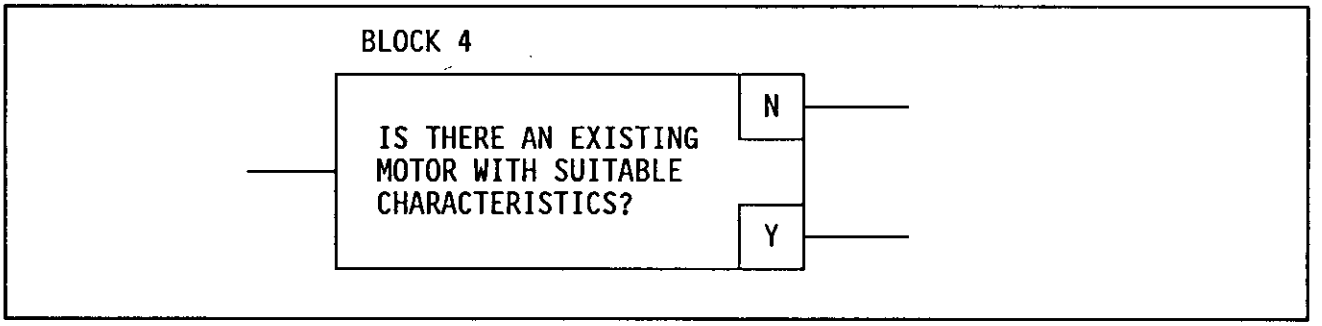
REFERENCES: Contract, System Design Engineer, System Design Team, Developmental Engineer, Logisticians, Motor Manufacturers.

BLOCK 3

PURPOSE: Block 3 is a command block. It requires that motor characteristics be erified before the motor selection process begins.

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5.3.3 Block 4. Existing motor.



BLOCK 4

PURPOSE: Block 4 focuses upon using selected motor characteristics to identify a suitable motor. It is at this time that an intelligent selection of a motor from existing designs can be made.

PRIMARY PARTICIPANT: Motor Designer.

SECONDARY PARTICIPANT(S): System Design Engineer (Mfr), System Design Team (Mfr), Developmental Engineer (User), Logisticians, Motor Manufacturers.

RESPONSIBILITIES: The Motor Designer is responsible for determining if an existing motor has all of the characteristics determined to be necessary for the new system's small electric motor. Secondary participants assist the Motor Designer as necessary.

DISCUSSION: A superficial check of design documentation and motor manufacturer catalogs may quickly provide a yes or no answer to the block 4 question. However, the answer should not be held to be final.

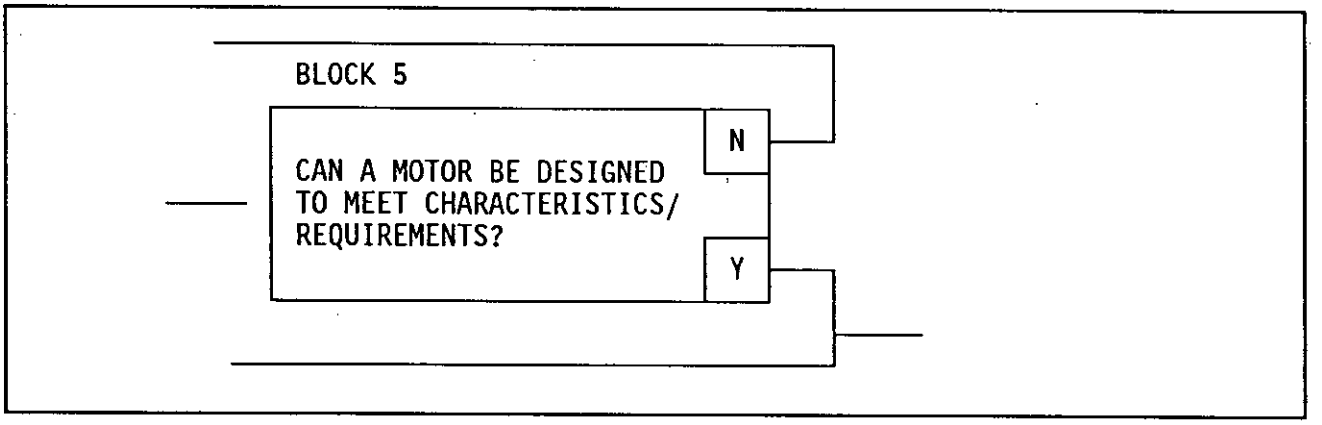
Further investigation should be made to ultimately arrive at the correct answer. Keeping in mind the problems discussed in Section 4, "Small Electric Motor Survey". motors currently available should be carefully analyzed for suitability. Manufacturer histories, product tests, and product performance should be obtained and evaluated. NEMA standard motors should be considered for first choice, followed by other commercial and federal specification type motors. Standard motors should be used if at all possible.

Keep in mind the need for easily obtained, replaceable, repairable motors. The User needs motors, and motors on back-order will not keep the new system in use. Logistics related inputs may influence the motor selection process.

REFERENCES: Motor characteristics; commercial, federal, military standards and specifications; design data; manufacturer's catalogs.

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5.3.4 Block 5. Motor design.



BLOCK 5

PURPOSE: The main purpose of block 5 is to focus on the problem of initially being unable to design a small electric motor when existing motors are unsuitable for use in the new system.

PRIMARY PARTICIPANT: Motor Designer.

SECONDARY PARTICIPANT(S): System Design Engineer (Mfr), System Design Team (Mfr), Developmental Engineer (User), Logisticians, Motor Manufacturers.

RESPONSIBILITIES: The Motor Designer is responsible for providing a satisfactory motor design for the new system. Secondary participants will assist the Motor Designer when necessary.

DISCUSSION: When a motor design meets its required characteristics it most likely can be produced as specified.

The problem of being unable to produce a satisfactory motor design involves two parameters: an incompetent motor designer, insufficient or inaccurate characteristics, or a combination of both. Assuming the motor designer is competent in that discipline, the problem becomes one of incompatibility between the motor design goal and the available characteristics.

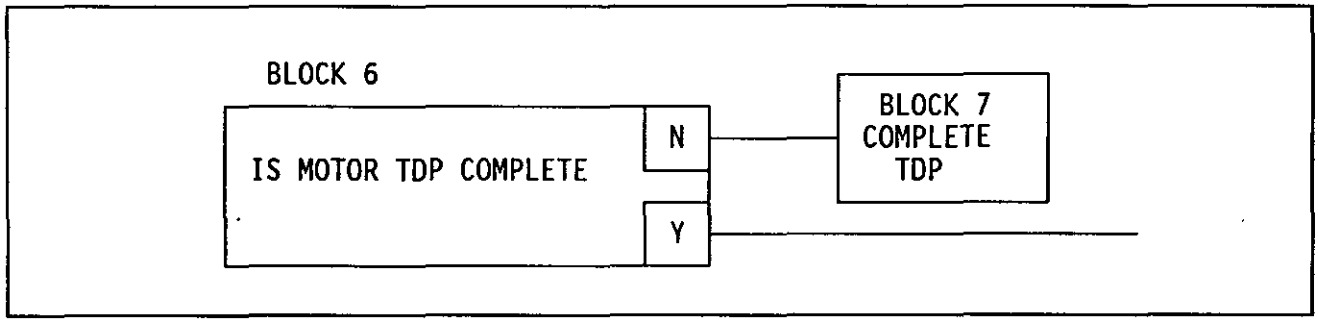
Faced with this situation, the Motor Designer must recheck all information and work which created the situation. Upon confirming the accuracy of work accomplished, the Motor Designer must then go back to the System Design Engineer, System Design Team, Developmental Engineer, Logisticians, and Motor Manufacturer(s) for additional guidance and consultation. This joint exercise should produce more, or more accurate characteristics for use in attaining the motor design goal. This activity most likely would take place back in block 2 of the decision network.

Once the design goal has been attained a new small electric motor specification can be prepared. It should address the same information as presented in block 2 (see paragraph 5.3.2).

REFERENCES: System specification system design; motor characteristics; motor specifications (all types); motor manufacturers.

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5.3.5 Block 6 and Block 7. Motor TDP.



BLOCK 6

PURPOSE: The purpose of block 6 is to place emphasis on the contents and use of a Technical Data Package (TDP).

PRIMARY PARTICIPANT: System Design Engineer (Mfr).

SECONDARY PARTICIPANT(S): Motor Designer, System Design Team (Mfr), Development Engineer (User), Logisticians, System Design Team.

RESPONSIBILITIES: The System Design Engineer is responsible for overall system development to include the selection of integral small electric motors, Working in conjunction with a Motor Designer and at times the Development Engineer and Logisticians, the System Design Engineer directs the development of a TDP for the system's small electric motor(s). During the TDP development process, the System Design Engineer must assure that, when necessary, all rights to technical data are obtained from appropriate sources.

DISCUSSION: The terms "Technical Data Package" and "Technical Data" are defined in paragraphs 3.58 and 3.59. Because of their importance these definitions are repeated here.

Technical Data Package.

"A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists, specifications, standards, models, performance requirements, quality assurance provisions, and packaging data."

Technical Data:

"Recorded information, regardless of form or characteristic, of a scientific or technical nature. It may, for example, document research, experimental, developmental, or engineering work; or be usable or used to define a design or process or procedure, produce, support, maintain, or operate materiel. The data may be graphic or pictorial delineations in media such as drawings or photographs, text in specification or related performance or design type documents, or computer

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printouts. Examples of technical data include research and engineering data, engineering drawings and associated lists, specifications, standards, process sheets, manuals, technical reports, catalog item identifications and related information, and computer software documentation. Technical data does not include computer software or financial, administrative, cost and pricing, and management data, or other information incidental to contract administration."

A military TDP consists of all the technical data needed to competitively procure, install, and maintain military material. Close attention to detail is required to develop a TDP which provides all information necessary for a potential manufacturer of an item to prepare and submit a responsive and responsible bid, quote, or proposal as solicited by the Government. Commercial TDP's contain similar data and are used for the same purposes.

The TDP is also used for establishing alternate sources of production for given items. Failure to include full and accurate data in such a TDP will assume poor performance by alternate source contractors.

Data to be included in a TDP are:

- Engineering drawings and associated lists.
- Product specifications.
- System and development specifications.
- Logistic Support data item descriptions for: provisioning technical documentation; technical manuals for operations, maintenance, installation, repair and replacement; and training.
- Special production, and training.
- Special test requirements to include component equivalency tests and tooling tests.
- Acceptance inspection equipment drawings and documentation.
- Packaging data sheets.
- Critical application designation (if necessary).
- Quality assurance provisions.
- Supplementary quality assurance data.
- Preparation for delivery instructions.
- Manufacturing and process descriptions.
- Ordering data.
- Identification and marking data.
- Specifications and Standards (commercial, federal, military).

After a TDP has been prepared, it must be checked for adequacy and accuracy. This effort includes checking for: adequate and accurate drawings, dimensions, tolerances, notes, circuitry, test procedures; completeness and accuracy of manufacturing data; adequacy of quality assurance provisions, adequacy of packaging and shipping data. Errors should be documented, and reasons for changes should be justified.

The importance of having an accurate, complete TDP cannot be overemphasized. An incomplete and inaccurate TDP will yield an unsatisfactory motor. It will also be wasteful in terms of time and dollars necessary to redo the TDP.

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REFERENCES:

Federal Acquisition Regulation (FAR), Part 27.
Defense Acquisition Circular #84-1, 1 March 1984, (Part 27.401).
MIL-T-60530A(MU), Technical Data Package for AMC Materiel.
DI-E-115B, Technical Data Package.
DI-E-1127, Technical Data Package Review Report.
DI-E-4062, Technical Data for Logistic Support.

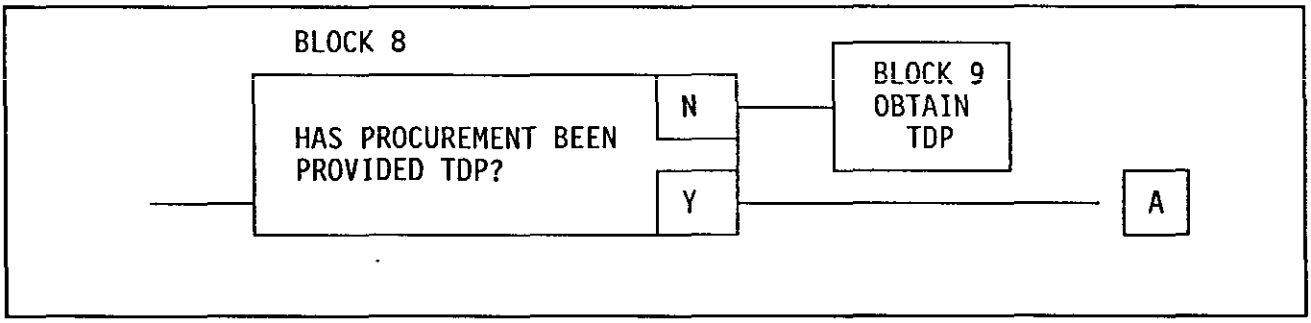
Technical documents discussed in paragraphs 4.4.1 and 4.4.2.

BLOCK 7

PURPOSE: Block 7 is a command block. It requires that a TDP be completed so that procurement can prepare a complete solicitation for the small electric motor. The "critical path" of this decision network cannot be completed without a TDP.

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5.3.6 Block 8 and Block 9. Procurement TDP.



BLOCK 8

PURPOSE: Block 8 addresses forwarding a TDP to the procurement office.

PRIMARY PARTICIPANT: System Design Engineer (Mfr).

SECONDARY PARTICIPANT(S): Contracting Officer (Mfr).

RESPONSIBILITIES: The System Design Engineer is responsible for forwarding the small electric motor TDP to the Contracting Officer. In turn, the Contractor Officer is responsible for confirming that the TDP has been received at the procurement office.

DISCUSSION: When the System Design Engineer forwards the small electric motor TDP to the Contracting Officer, the bridge between the requirements activity and the procurement activity is crossed. Failure to accomplish this task on time can delay the award of a contract for the motor. This delay can, in turn, impact downstream production and delivery schedules. Being aware of the schedule, the Contracting Officer must know if the TDP was received by procurement according to schedule, and is being properly processed by the procurement staff. Processing the TDP includes checking it for conformance to the format and contents required by local and customer directives.

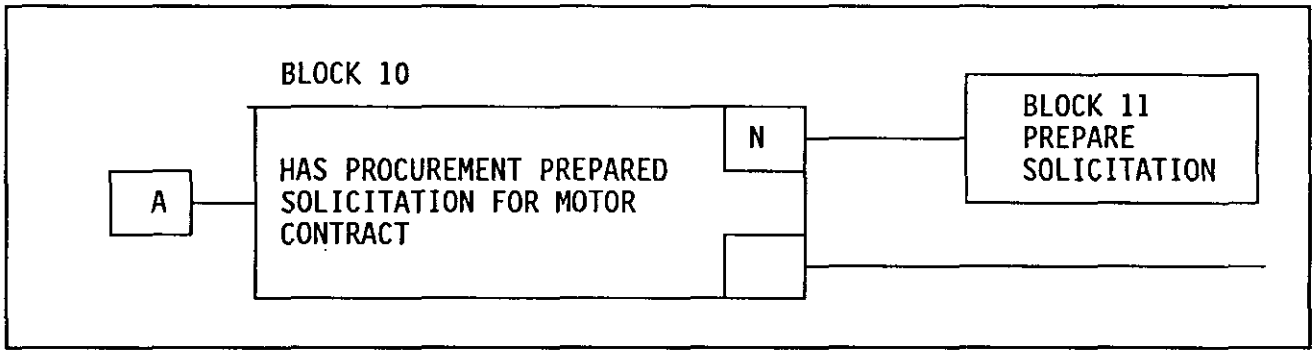
REFERENCES: Manufacturer's directives concerning preparation, transmission, and receipt of TDPs.

BLOCK 9

PURPOSE: Block 9 is a command block. It requires that the Contracting Officer be aggressive and obtain the small electric motor TDP from the System Design Engineer.

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5.3.7 Block 10 and Block 11. Prepare solicitation.



BLOCK 10

PURPOSE: The purpose of block 10 is to highlight the importance of preparing the small electric motor solicitation according to schedule.

PRIMARY PARTICIPANT: Contracting Officer (Mfr).

SECONDARY PARTICIPANT(S): System Design Engineer (Mfr), Motor Designer.

RESPONSIBILITIES: The Contracting Officer is responsible for preparing the small electric motor solicitation. Secondary participants support the Contracting Officer during the solicitation preparation effort.

DISCUSSION: The solicitation for the small electric motor must contain all details necessary to clearly describe what is wanted, when it is wanted, how it is wanted, and why it is wanted. The more data and detail in the solicitation package, the better the response from a potential small electric motor manufacturer. A well prepared solicitation will almost always yield sufficient positive responses which, in turn, show that an industrial base exists for a given item.

Before a solicitation is prepared, the Contracting Officer must decide whether or not to solicit small business firms. At times a military contract may require that small businesses be solicited. The decision is based upon prime contract requirements and company guidelines for small business procurements.

If the decision is for a small business solicitation, then another decision is required: Whether or not to solicit labor surplus areas and minority small businesses? Again, the decision may be influenced by the prime contract and company guidelines.

Timely preparation of the small electric motor solicitation should result in timely attainment of subsequent steps in acquiring the motor. To this end, the Systems Design Engineer and Motor Designer should assist the Contracting Officer in seeing that the solicitation is prepared on time.

In the event that the Contracting Officer does not need assistance, the System Design Engineer should track the solicitation to assure it goes out on time. Time lost to procurement cannot always be made up by engineering shortcuts.

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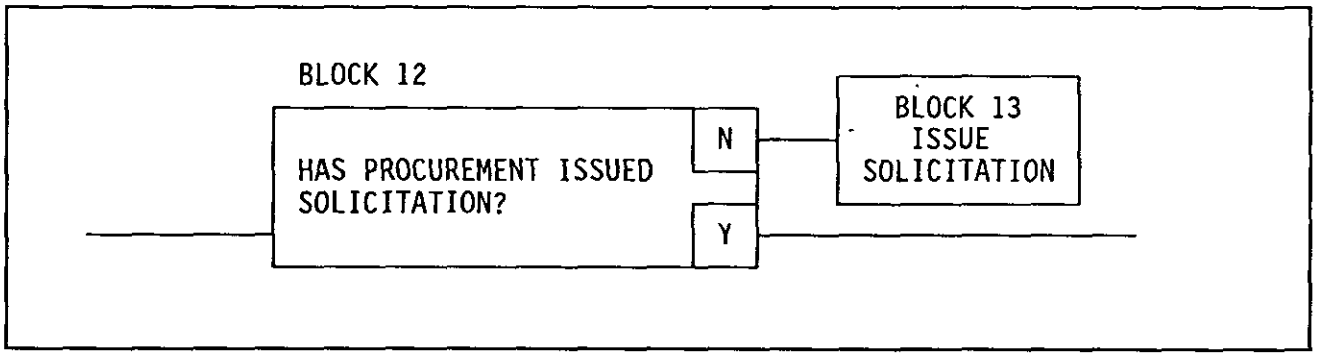
REFERENCES: Manufacturer's procurement guidelines, directives, regulations.

BLOCK 11

PURPOSE: Block 11 is a command block. It requires that the small motor solicitation be prepared if not already done so.

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5.3.8 Block 12 and Block 13. Issue solicitation.



BLOCK 12

PURPOSE: The purpose of block 12 is to establish a check point for determining if the small electric motor solicitation has been issued.

PRIMARY PARTICIPANT: Contracting Officer (Mfr).

SECONDARY PARTICIPANT(S): System Design Engineer (Mfr).

RESPONSIBILITIES: The Contracting Officer is responsible for issuing the small electric motor solicitation. The System Design Engineer should provide full cooperation and assistance as necessary.

DISCUSSION: This seemingly inconsequential check point represents the bridge between a prime contractor and those subcontractors who supply contract end articles for use in the prime's end item. The bridge cannot be crossed until the solicitation is "on-the-street".

Solicitation schedules are part of the overall new system schedule. Both the Contracting Officer and the System Design Engineer should be concerned for assuring the timely issuance of new system hardware solicitations. Failure to adhere to the solicitation issuance schedule can adversely impact downstream system production and delivery activities. The System Design Engineer should anticipate the need to provide assistance to the Contracting Officer if so requested.

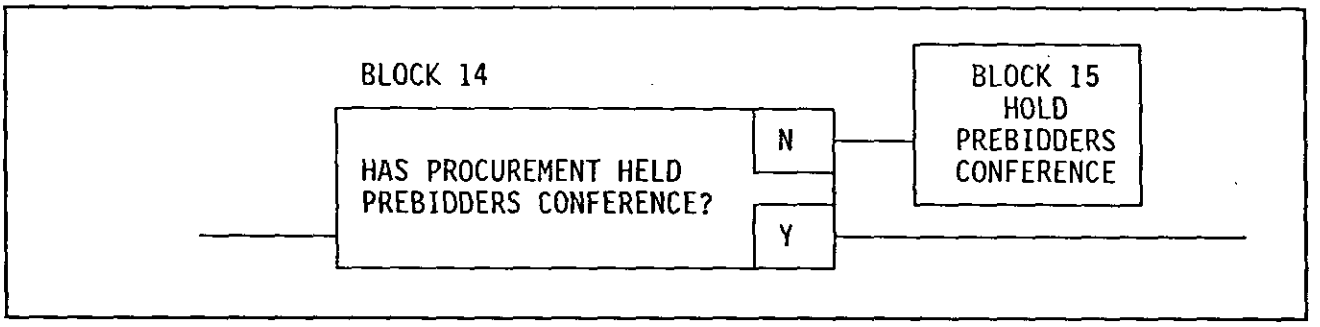
REFERENCES: Manufacturer's procurement guidelines, directives, regulations.

BLOCK 13

PURPOSE: Block 13 is a command block. It requires that a solicitation be issued if not already done so.

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5.3.9 Block 14 and Block 15. Prebidders conference.



BLOCK 14

PURPOSE: Block 14 concerns a prebidders conference for the new system's small electric motor.

PRIMARY PARTICIPANT: Contracting Officer (Mfr).

SECONDARY PARTICIPANT(S): System Design Engineer (Mfr), Motor Designer, Potential small electric motor suppliers.

RESPONSIBILITIES: The Contracting Officer is responsible for assuring that potential small electric motor manufacturers have the opportunity to jointly attend a conference concerning the solicitation. Engineering staff members such as the Motor Designer and System Design Engineer should attend this conference to supply technical information.

DISCUSSION: A prebidders conference is not always required. But, by scheduling and holding one, the Contracting Officer and secondary participants can easily see who the potential suppliers are, and what their understanding of the solicitation is.

During a prebidders conference, potential suppliers have the opportunity to ask questions and obtain responses. Invariably, a transfusion of information occurs, and the suppliers gain a better understanding of the solicitation. At times, a solicitation is amended due to the need to insert clarifying statements within.

Some suppliers lose interest, while others decide to compete for the contract award. Discussions cover wide range of topics, including those for:

- Availability of contractor and government specifications and standards.
- Test data and reports.
- Special test requirements.
- Use of end article (multiple use?).
- Maintenance requirements.
- Preaward surveys.
- Quality assurance.
- Substitute items.
- Former government surplus items.
- Schedules
- Payments, progress payments.
- Proof tests.

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- First article qualification tests.
- Qualification tests.
- Schedules.
- Prime contractor and government contract administration.

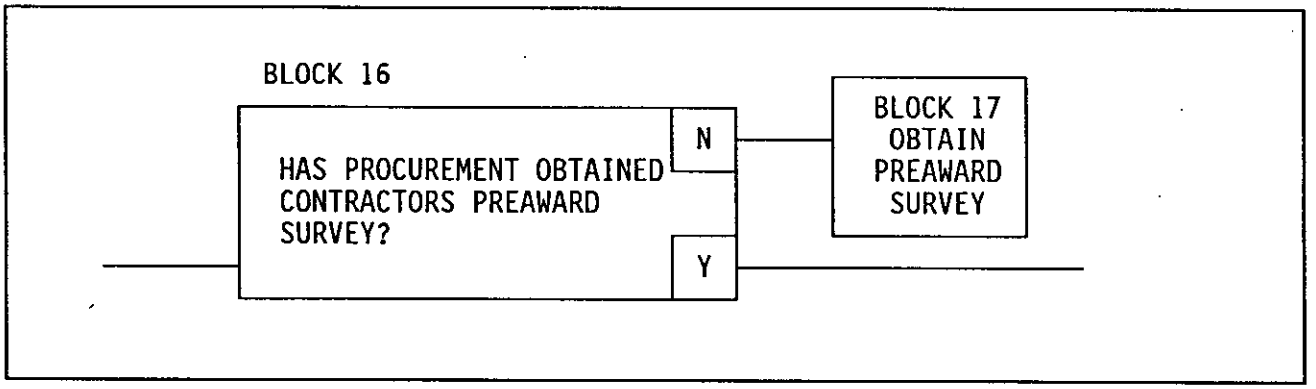
REFERENCES: Manufacturer's procurement guidelines, directives, regulations.

BLOCK 15

PURPOSE: Block 15 is a command block. It reinforces the desirability of holding a prebidders conference.

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5.3.10 Block 16 and Block 17. Preaward survey.



BLOCK 16

PURPOSE: The purpose of block 16 is to provide information about preaward surveys.

PRIMARY PARTICIPANT: Contracting Officer (Mfr).

SECONDARY PARTICIPANT(S): Motor Designer, Production Engineer (Mfr), Quality Engineer (Mfr), System Design Engineer (Mfr).

RESPONSIBILITIES: The Contracting Officer is responsible for conducting a preaward survey. Secondary participants join in the survey, either by directive or by choice.

DISCUSSION: A preaward survey is conducted for the purpose of aiding the Contracting Officer in making the decision as to which bidder will be finally selected for any given contract award. By carefully reviewing bidder's inputs, and making use of all available information, the Contracting Officer can usually narrow the number of responsive and responsible bidders down to the single one most likely to be able to complete the contract satisfactorily. When confirmation is desired or required, the Contracting Officer initiates the preaward survey process.

Some preaward surveys can be accomplished by a "desk audit". This action usually covers those suppliers who have constantly produced satisfactory end articles for use by their customers. A check on plant loading may be all that is required for confirmation of a decision to award a contract to that supplier.

On the other end of the spectrum is the need for a complete preaward survey. For whatever reason, the Contracting Officer wants an in-plant check of the potential contract winner. Items to be checked include plant management, plant administration, financial stability and history, present debt, plant loading, product history, product quality assurance, delivery schedules, adequate and properly trained work force, and current production of the same or similar items.

The Contracting Officer designates experienced persons to check each area of interest. Other persons may request to participate in the survey. Each input rates the potential supplier as to being capable and responsive to contract requirements. Finally, each preaward designated participant makes a recommendation, fully justified, as to whether or not the potential supplier should be awarded the

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contract. This information is used by the Contracting Officer to make the final yes or no decision concerning the contract award.

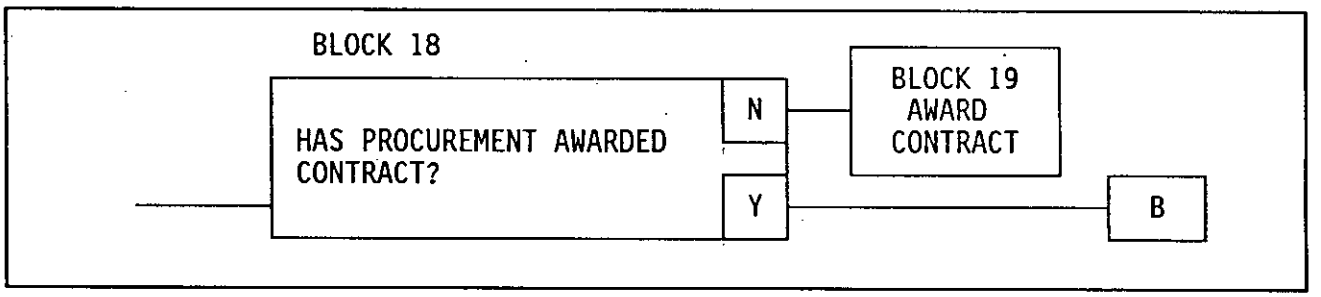
REFERENCES: Manufacturer's subcontractor survey guidelines, directives, regulations.

BLOCK 17

PURPOSE: Block 17 is a command block. It requires that a preaward survey be used to assist the Contracting Officer in making contract award decisions.

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5.3.11 Block 18 and Block 19. Contract award.



BLOCK 18

PURPOSE: The purpose of block 18 is to highlight the preparation and award of the small electric motor contract.

PRIMARY PARTICIPANT: Contracting Officer (Mfr.)

SECONDARY PARTICIPANT(S): Motor Manufacturer.

RESPONSIBILITIES: The Contracting Officer is responsible for developing a contract to be offered to the Motor Manufacturer. The Motor Manufacturer is responsible for understanding the terms and conditions of the contract, and for accepting or otherwise coming to an agreement with the Contracting Officer.

DISCUSSION: A contract is the visible result of an offer, acceptance, and mutual consideration. Put another way, a contract represents a combination of (1) form, (2) compensation arrangement, and (3) end purpose, e.g., a purchase order (form) with a fixed price (compensation arrangement) for supplies (end purpose).

The award of the contract is the end result of a lot of preparatory activity by the primary and secondary participants. For the Contracting Officer, blocks 8-17 set the stage for contract award. For the winning motor manufacturer, the activity begins when the solicitation is received.

Both participants must work to their internal schedules: system manufacturer - system delivery schedule; motor manufacturer - motor delivery schedule. Cost and pricing data is developed by both participants. The motor manufacturer's price is based upon costs plus a profit margin. This price is reviewed by the system manufacturer, and a price analysis may be performed to determine the fairness of the price. Finally, when all details are agreeable to both participants, the contract is completed and awarded to the motor manufacturer.

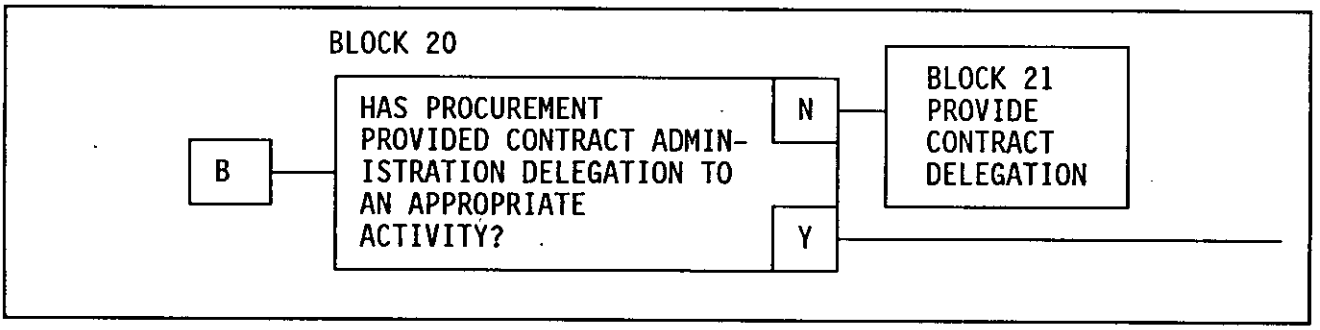
REFERENCES: System Manufacturer's and Motor Manufacturer's guidelines, directives, regulations.

BLOCK 19

PURPOSE: Block 19 is a command block. It serves as a check point to assure a contract is awarded within schedule requirements.

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5.3.12 Block 20 and Block 21. Contract administration.



BLOCK 20

PURPOSE: The purpose of block 20 is to provide a brief on contract administration.

PRIMARY PARTICIPANT: Contracting Officer (Mfr).

SECONDARY PARTICIPANT(S): Motor Manufacturer, DCAS.

RESPONSIBILITIES: The Contracting Officer is responsible for administration of the small electric motor contract at the prime contractor-subcontractor level of activity. Similarly, the Motor Manufacturer is responsible for contract administration at the motor manufacturing plant and external supplier level of activity. Government contract administration focuses principally upon the prime contractor, and at times on a prime contractor's subcontractor(s).

DISCUSSION: The contract administration function is the focus of all activity between contracting parties. Some of the many facets of contract administration are:

- Contract amendments or changes.
- Subcontracts and purchase orders.
- Cost and price analysis.
- Property control.
- Production control.
- Quality assurance, quality control.
- Subcontractor proposal reviews.
- Subcontract negotiations.
- Labor law compliance.
- Correspondence and reports.
- Allowable costs.
- Safety.
- Security.

At the prime level the Contracting Officer may perform contract administration functions with an internal staff, or by delegation to other functions of the system manufacturer; i.e., quality control, production, shipping and receiving, and subcontractor surveillance. Subcontractors may perform similar functions at their manufacturing plant(s).

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Government contract administration is usually limited to those contractors having government contracts. But, the government retains the right to check any or all steps of manufacturing of its material, with or without privity of contract. Arrangements for government inspection are made by both the prime contractor - subcontractor route and government - subcontractor route. In the latter case, the government contracting office delegates the inspection of the subcontractor to the Defense Contract Administration Services (DCAS) or some other government contract administration function. DCAS, for example, will then notify the subcontractor of the need for government inspection. The prime contractor also notifies the subcontractor of the government intent. Subsequent to these notifications, the government then performs its inspections as necessary.

Separate inspections of subcontractors by both the prime contractor and the government usually assure that acceptable products are delivered to the government. Product deficiencies that occur may be traced to:

- Poor TDP's.
- Insufficient inspections and tests.
- Unskilled inspectors.
- Inadequate production controls.
- Unskilled work force.
- Awards to incompetent manufacturer.
- Poor preaward surveys.

Proper feedback of contract and contractor deficiencies for review and correction by contract administration activities should reduce both future problems, occurrences, and receipt of deficient products for government use.

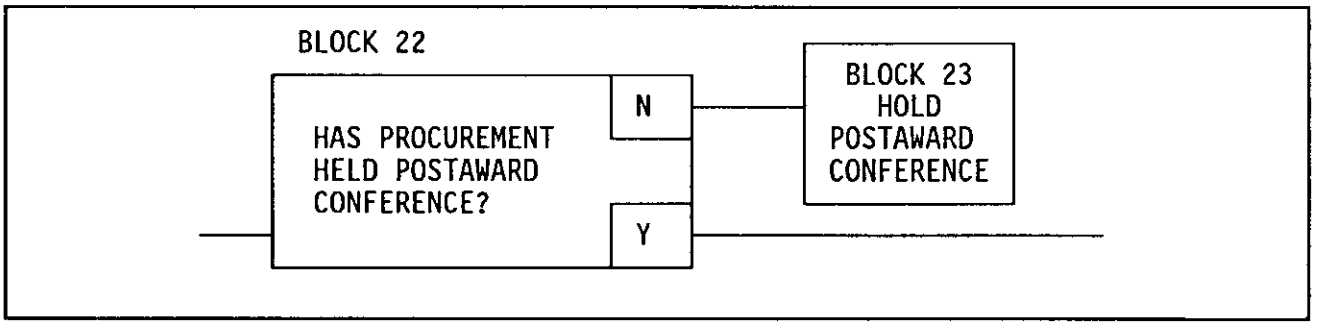
REFERENCES: System manufacturer's guidelines, directives, regulations; contract requirements.

BLOCK 21

PURPOSE: Block 21 is a command block. It requires that contract delegations be made for field surveillance of the motor manufacturer's contract performance.

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5.3.13 Block 22 and Block 23. Postaward conference.



BLOCK 22

PURPOSE: The purpose of block 22 is to identify the postaward conference function and provide some information of its activity.

PRIMARY PARTICIPANT: Contracting Officer (Mfr).

SECONDARY PARTICIPANT(S): System Design Engineer, Motor Designer, Motor Manufacturer, DCAS, other contract administration activity.

RESPONSIBILITIES: The Contracting Officer is responsible for making arrangements and holding the small electric motor contract postaward conference. DCAS or another government contract administration activity may attend by request or by choice. Other secondary participants are parties to the conference.

DISCUSSION: During the postaward conference, all participants discuss the small electric motor contract, its requirements to include schedules, and the functions and responsibilities of each party. Questions are raised and answered, and procedural problems are resolved. Surveillance actions by both the Contracting Officer and DCAS, or other designated contract administration activity, are explained as necessary to the Motor Manufacturer.

The results of this conference should leave no question unanswered concerning the function, conduct, and mutual and individual responsibilities for successful contract performance.

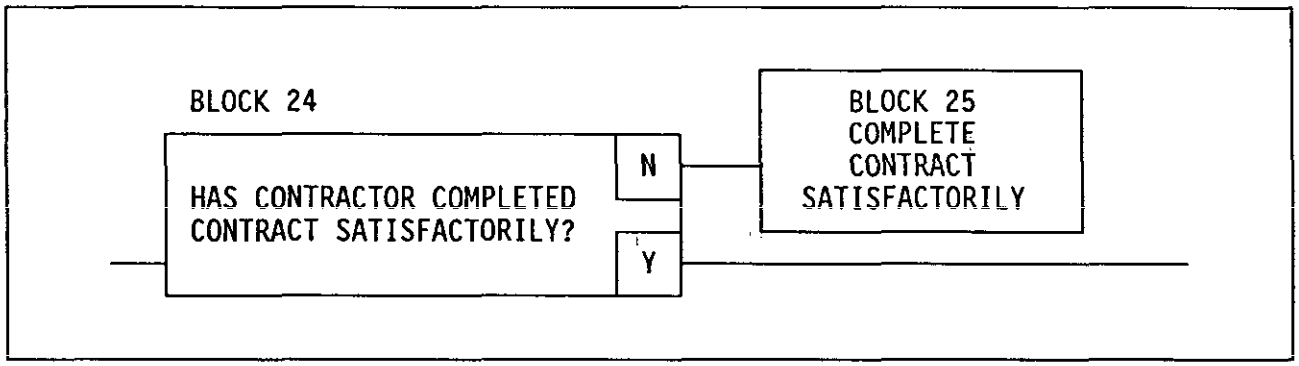
REFERENCES: System manufacturer's guidelines, directives, regulations; contract requirements; contract delegation.

BLOCK 23

PURPOSE: Block 23 is a command block. It requires that a postaward conference be held, preferably before the motor manufacturer commences work.

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5.3.14 Block 24 and Block 25. Contract completion.



BLOCK 24

PURPOSE: The purpose of block 24 is to highlight the satisfactory completion of a contract.

PRIMARY PARTICIPANT: Motor Manufacturer.

SECONDARY PARTICIPANT(S): Contracting Officer (Mfr), DCAS, other contract administration activity.

RESPONSIBILITIES: The Motor Manufacturer is responsible for the satisfactory completion of the small electric motor contract. Secondary participants are responsible for tracking the Motor Manufacturer's progress through contract completion.

DISCUSSION: The satisfactory completion of a contract involves more than customer acceptance of contract and articles. Depending upon contract requirements, the customer will almost always require other contract deliverables such as drawings, TDP's, reports, inspection data, technical manuals, and financial and progress reports.

Customer in-process hardware inspections at its subcontractor's facility are one of the most important steps in assuring satisfactory contract completion. These inspections should provide data that confirms satisfactory performance by the subcontractor. One caution at this point is that the customer must not relax or reduce its inspections due to concurrent DCAS inplant inspections (see block 21). Both customer data and independently supplied government data should be used to determine whether or not the subcontractor is performing satisfactorily.

Each contract deliverable has a time and place for delivery and customer acceptance. Regardless of when and where scheduled deliveries and acceptances occur a contract will not be satisfactorily completed unless all items are delivered on time and accepted by the customer at that time.

Subcontractor milestone delays may be costly to the subcontractor, if due to poor performance, in terms of penalty payments imposed by the contract. And, each unsatisfactorily completed contract milestone becomes, or should be a static for the customer's contractor performance file. These unsatisfactory actions may require

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contractual adjustment before a contract can be considered as being completed satisfactorily.

Finally, all contract participants should work together as much as possible to assure the satisfactory completion of a contract. Early identification and correction of problems almost always will prevent or reduce downstream contractual delays which adversely impact satisfactory contract completion.

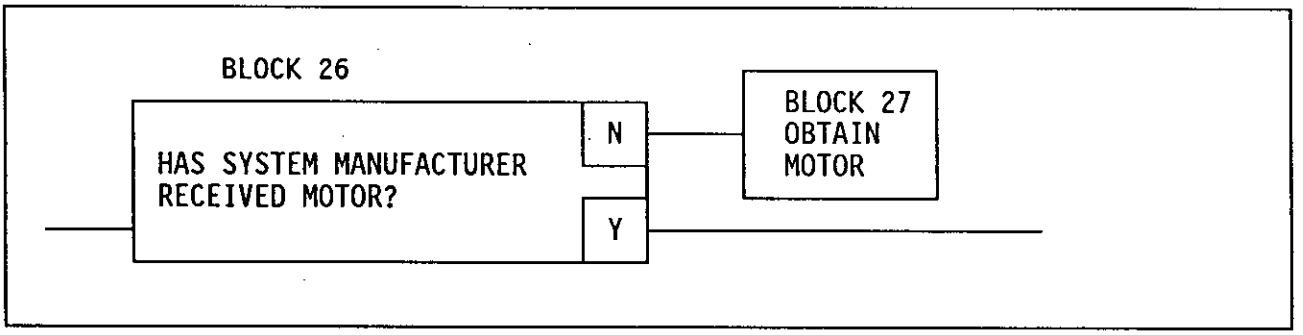
REFERENCES: Contract requirements; manufacturer's guidelines, directives, regulations, DCAS and other field activity guidelines, directives, regulations.

BLOCK 25

PURPOSE: Block 25 is a command block. It requires all contract requirements to be met for a contract to be considered satisfactorily completed.

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5.3.15 Block 26 and Block 27. Motor receipt.



BLOCK 26

PURPOSE: The purpose of block 26 is to direct attention to the receipt of an end article at a prime contractor's manufacturing facility.

PRIMARY PARTICIPANT: System Manufacturer.

SECONDARY PARTICIPANT(S): Motor Manufacturer.

RESPONSIBILITIES: The System Manufacturer is responsible for the receipt and accountability of the small electric motor. The Motor Manufacturer must ship the motor as directed by the contract.

DISCUSSION: Shipping the new system's small electric motor from the Motor Manufacturer is just the beginning of the motor receipt process.

The Motor Manufacturer must ship the motor according to contract requirements. Usually a contract will stipulate levels of packaging and packing, mode of transportation, destination, and required receipt date. Part of the shipping process may include notifying the System Manufacturer of the shipment date and the name of the cargo carrier.

Upon receipt of notification of motor shipment, the System Manufacturer's receiving department prepares for the motor's arrival. Receipt procedures should include inspection of the motor shipment for count and condition of packaging and packing. If there is a shipment problem related to motor damage, it has to be resolved by the carrier, motor manufacturer, and system manufacturer.

Following receipt of the motor, there may be the requirement for bench tests and storage until needed for installation in the new system. Also, concurrent inspection of accompanying technical data, such as technical manuals and handbooks, should be accomplished. All discrepancies should be resolved by actions of the system manufacturer and motor manufacturer.

Finally, all system manufacturer departments concerned with the small electric motor must be notified of its arrival, status, and location.
REFERENCES: Contract requirements; Motor Manufacturer and System Manufacturer guidelines, directives, regulations.

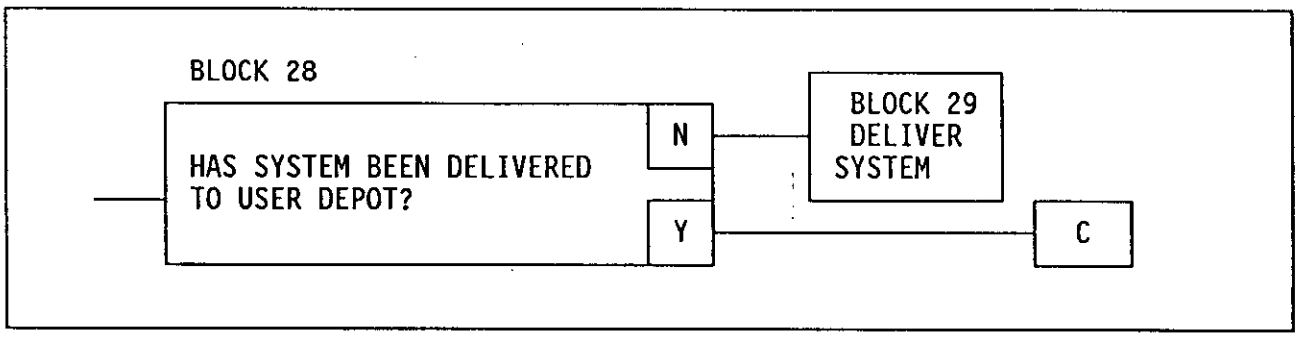
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BLOCK 27

PURPOSE: Block 27 is a command block. It requires the System Manufacturer to take aggressive action to see that the motor is on hand when required.

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5.3.16 Block 28 and Block 29. System delivery.



BLOCK 28

PURPOSE: Block 28 represents the bridge between the new system manufacturer, small electric motor manufacturer, and the new system user.

PRIMARY PARTICIPANT: User Depot.

SECONDARY PARTICIPANT(S): System Design Engineer (Mfr), DCAS.

RESPONSIBILITIES: The User Depot is responsible for the receipt, inspection, acceptance is required, storage, and subsequent issuance of the new system, with its small electric motor, to the User. Secondary participants are responsible, in whole or in part, for system manufacture, tests and checkouts, inspection, acceptance, and shipping to the depot or User.

DISCUSSION: The small electric motor has been installed, and the new system has been completed, tested, inspected, and shipped of the User Depot. Depot records must be initiated or updated to show delivery of the new system.

Part of the delivery process includes the assurance that previously provisioned spare and repair parts are now stocked at their designated sites. Operations and technical manuals should also be available for both depot and User personnel. Authorized tools and test equipment have been established according to the previously established maintenance concept. (See block 1 and block 2).

While at the depot, the new system could require a later modification as part of the delivery process. Such a modification could be done by both depot and secondary participants utilizing depot facilities. If modifications were not necessary, the new system may have been directly shipped to the User--bypassing the depot.

Storage facilities are set aside for the new system and its spare and repair parts Environmental control conditions have been established for the system and its equipment. This includes temperature, humidity control, and protection from airborne contaminants, such as dust and salt sprays.

Maintenance data collection is very important for depot maintenance of the new system. Trained personnel should be in place and provisions for data collection should be established before the new system is delivered to the User.

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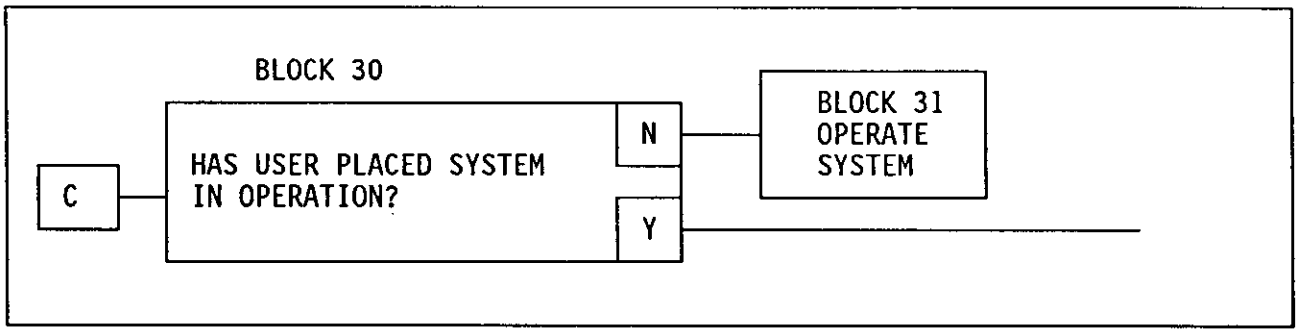
REFERENCES: Military department guidelines, directives, and regulations; System Manufacturer and DCAS guidelines, directives, and regulations.

BLOCK 29

PURPOSE: Block 29 is a tollgate. It requires that the new system be delivered, including all items necessary for system support, before the system is transferred from the depot to the User.

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5.3.17 Block 30 and Block 31. Operate system.



BLOCK 30

PURPOSE: The purpose of block 30 is to focus attention upon the beginning of operational use of the new system and its small electric motor.

PRIMARY PARTICIPANT: User.

SECONDARY PARTICIPANT(S): User Maintenance Activity, User Depot, System Manufacturer.

RESPONSIBILITIES: The User is responsible for achieving system initial operational capability. Secondary participants support the User as necessary.

DISCUSSION: The small electric motor, as part of the new system, has started its operational life. To achieve this step the User had to prepare facilities for the new system; obtain technical manuals, tools, and test equipment; train assigned operations and maintenance personnel; and establish stock levels for spare and repair parts.

System Manufacturer technical assistance may be required during the beginning stages of achieving system initial operational capability. This assistance may be part of an existing contractual arrangement, or it may be obtained by initiating a new agreement. Regardless of how obtained, there should be no delays in getting such assistance when required.

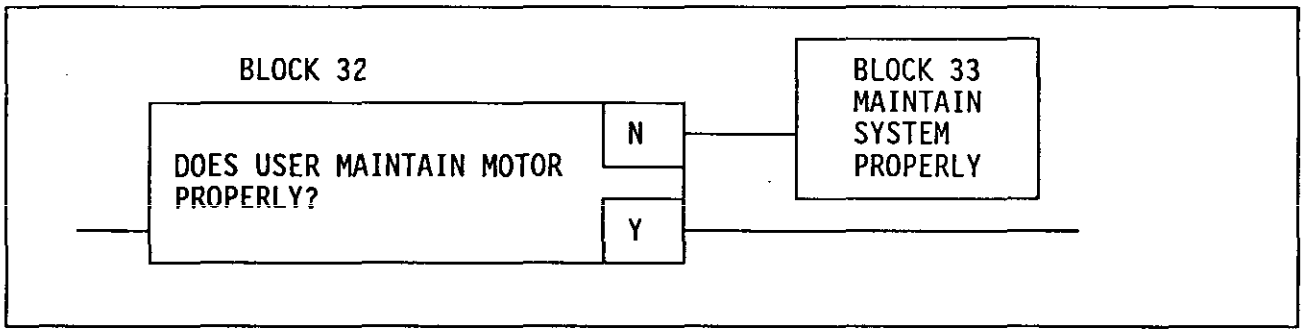
REFERENCES: Departmental guidelines, directives, regulations.

BLOCK 31

PURPOSE: Block 31 is a command block. It requires that the User and support elements complete all actions necessary to attain initial operational capability and to sustain full operations from then on.

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5.3.18 Block 32 and Block 33. Motor maintenance.



BLOCK 32

PURPOSE: Block 32 points out that the new system's small electric motor must be maintained according to its previously established maintenance concept (see block 1 and block 2).

PRIMARY PARTICIPANT: User.

SECONDARY PARTICIPANT(S): User Maintenance Activity, User Depot.

RESPONSIBILITIES: The User is responsible for correctly maintaining assigned systems and equipment. The User Maintenance Activity and User Depot support the User as necessary.

DISCUSSION: Small electric motors almost always require some preventive maintenance activity. This ranges from simple exterior surface cleaning to inspection and lubrication of bearings.

As part of the new system, the small electric motor must be maintained according to established manufacturer and User maintenance instructions and procedures. Planned preventive maintenance for the small electric motor will contribute greatly to the motor meeting its design reliability requirements.

Failure to accomplish preventive and planned maintenance actions will almost always result in premature motor failures.

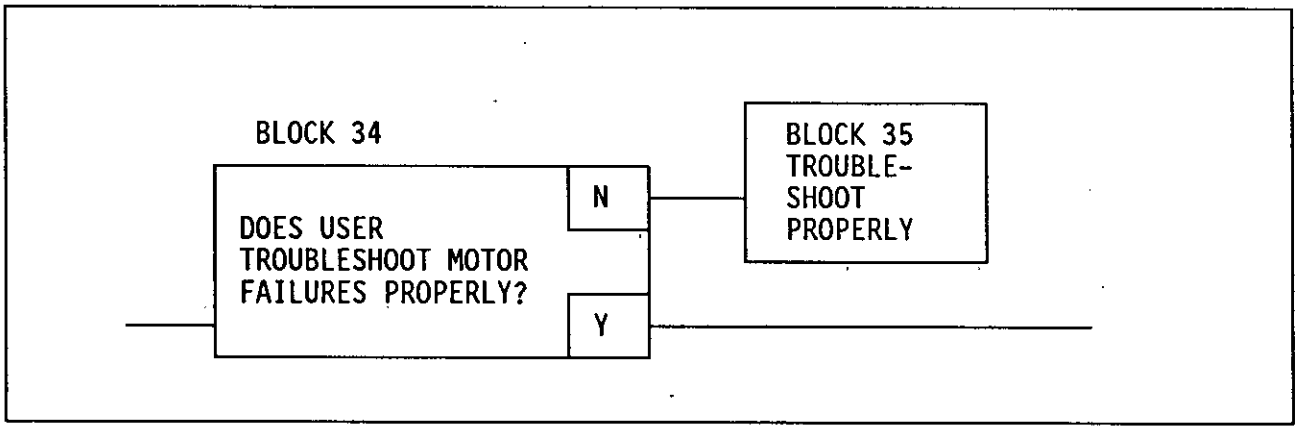
REFERENCES: Departmental guidelines, directives, regulations; manufacturer instructions; maintenance manuals.

BLOCK 33

PURPOSE: Block 33 is a command block. It requires that the User follow all established procedures for correct maintenance of the small electric motor.

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5.3.19 Block 34 and Block 35. Motor failures.



BLOCK 34

PURPOSE: Block 34 highlights troubleshooting the small electric motor according to established procedures.

PRIMARY PARTICIPANT: User Maintenance Activity.

SECONDARY PARTICIPANT(S): User Depot.

RESPONSIBILITIES: The User Maintenance Activity is responsible for troubleshooting inoperable equipment according to established maintenance procedures. The User Depot supports the User Maintenance Activity as necessary.

DISCUSSION: Well planned and formalized troubleshooting procedures provide a logical and systematic method for determining failure modes of systems and equipment. When trained and disciplined maintenance personnel use correct procedures and proper tools and test equipment, they will almost always find a failure problem without undue waste of time.

As the new system matures in use, these procedures may be modified based upon experience and introduction of new test and repair equipment.

Regardless of the complexity of an item, troubleshooting must be accomplished by following established procedures and using recommended test equipment that is properly maintained and calibrated.

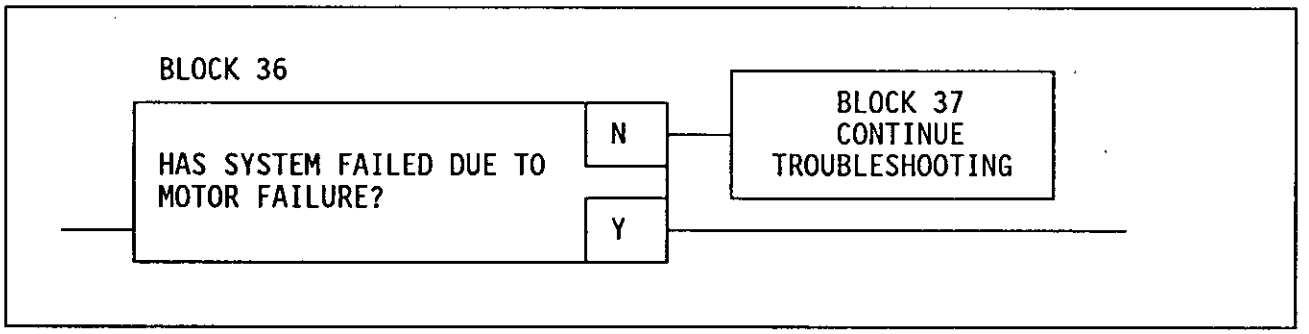
REFERENCES: Departmental guidelines, directives, regulations; manufacturer instructions; maintenance manuals.

BLOCK 35

PURPOSE: Block 35 is a command block. It requires that the use of poor procedures and practices be corrected to conform to established requirements. Supervisors at all levels must assure that corrective actions are proper.

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5.3.20 Block 36 and Block 37. System failure.



BLOCK 36

PURPOSE: The purpose of block 36 is to address the need to correctly troubleshoot and determine the cause of a system failure.

PRIMARY PARTICIPANT: User Maintenance Activity.

SECONDARY PARTICIPANT(S): User Depot.

RESPONSIBILITIES: The User Maintenance Activity is responsible for correctly determining failure causes of systems and equipment. The User Depot assists the User Maintenance Activity as necessary.

DISCUSSION: By following correct troubleshooting procedures, the new system failure has been traced to the failure of the small electric motor. The next action is to restore the system to operational use. This action can be achieved by repairing the motor locally, replacing the motor from local stocks, or obtaining a replacement motor from the depot.

Part of the system restoration process includes correctly identifying the motor's stock number and part umber, and recording discrepancy data for use by the item manager(s) and maintenance managers. At times, maintenance actions and item design are changed based upon the collection of maintenance data. Also, for new systems there may be warranty provisions for given items. Again, maintenance data may influence User claims against item manufacturers.

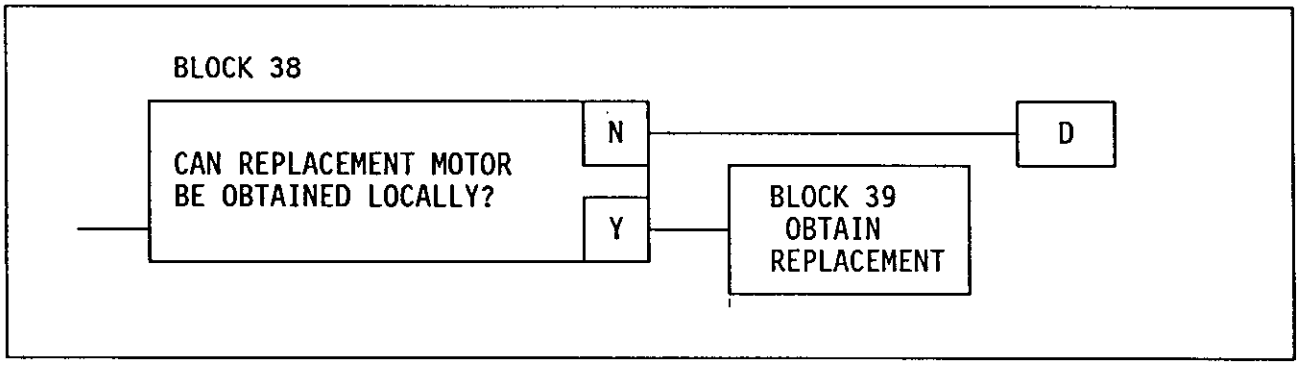
REFERENCES: Departmental guidelines, directives, regulations; manufacturer's instructions; maintenance manuals.

BLOCK 37

PURPOSE: Block 37 is a command block. It requires that the cause of the system failure be determined. This block recognizes that something other than our small electric motor has failed -- go find the failure cause and correct it.

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5.3.21 Block 38 and Block 39. Obtaining spare motor.



BLOCK 38

PURPOSE: The purpose of block 38 is to show some of the actions required for obtaining a spare motor.

PRIMARY PARTICIPANT: User Maintenance Activity.

SECONDARY PARTICIPANT(S): User Supply Activity, User Depot Supply Activity.

RESPONSIBILITIES: The User Maintenance Activity is responsible for obtaining a spare small electric motor from the User Supply Activity. Assistance from the User Depot Supply Activity is available as necessary.

DISCUSSION: The User Supply Activity will stock and issue spare motors for subsequent replacement in the new system by the User Maintenance Activity. Inoperable motors will be returned to, and disposed of, according to established supply activity instructions and the system's maintenance concept.

If there are no spares on hand, the User Supply Activity and the User Maintenance Activity may have to consider repairing motors, if authorized, at the User Maintenance Activity.

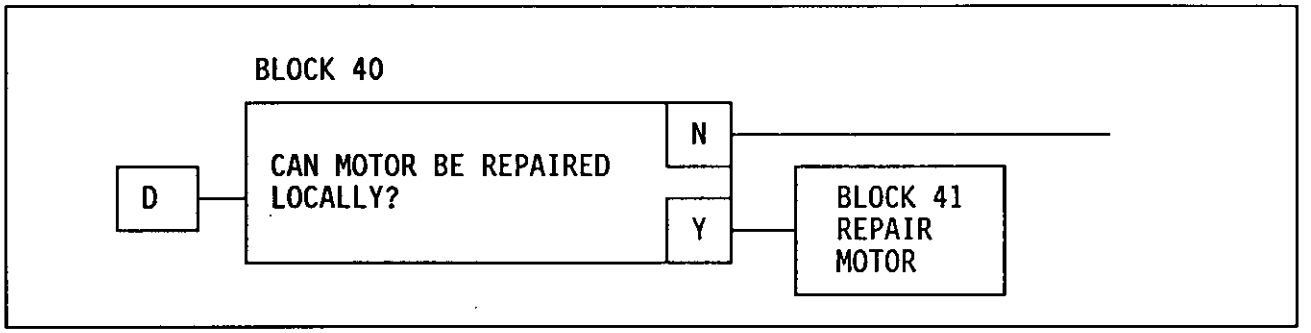
REFERENCES: Departmental guidelines, directives, regulations.

BLOCK 39

PURPOSE: Block 39 is a command block. It requires that a spare motor is issued by the User Supply Activity and is replaced by the User Maintenance Activity. The system is now operational.

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5.3.22 Block 40 and Block 41. Repairing motor locally.



BLOCK 40

PURPOSE: The purpose of block 40 is to address local repair of the new system's small electric motor.

PRIMARY PARTICIPANT: User Maintenance Activity.

SECONDARY PARTICIPANT(S): User Supply Activity, User Depot Supply Activity.

RESPONSIBILITIES: The User Maintenance Activity is responsible for repairing the small electric motor if feasible and authorized. The User Depot Supply Activity assists the User Supply Activity as necessary.

DISCUSSION: Establishing a local repair capability for systems and equipment is the result of earlier planning to determine the levels and extent of maintenance and repair actions. As a system and its components mature, the expansion of a local repair capability may occur.

Small electric motor maintenance actions must be continually managed to assure that maintenance (repair, overhaul, rebuild) is performed at the lowest possible and feasible level. Unapproved maintenance actions must, as a rule, be prohibited because of the possibility of an adverse impact upon established supply levels and manpower requirements at higher maintenance echelons.

If the small electric motor cannot be repaired locally within a previously established time frame, then the User Supply Activity must take action to obtain spares from the User Depot Supply Activity.

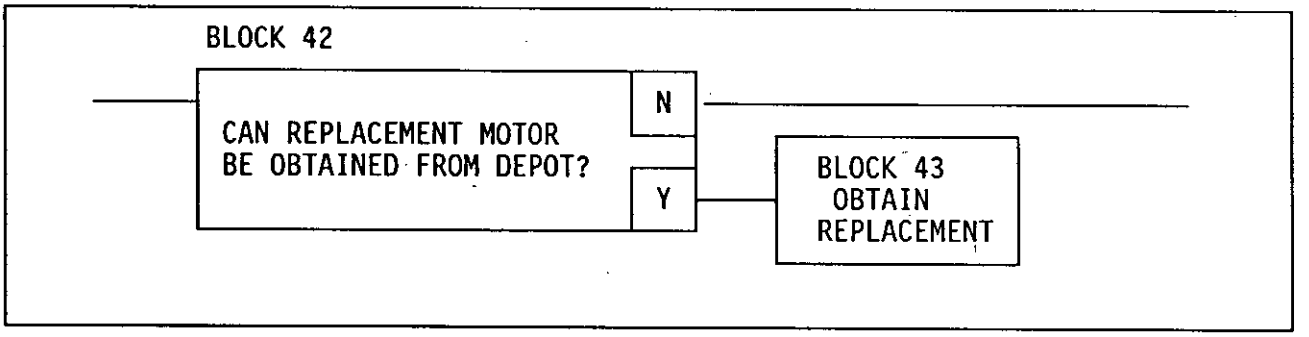
REFERENCES: Departmental guidelines, directives, regulations; manufacturer's instructions; maintenance manuals.

BLOCK 41

PURPOSE: Block 41 is a command block. It requires that the small electric motor is repaired locally according to approved practices and procedures by utilizing approved parts and test equipment. The system is now operational.

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5.3.23 Block 42 and Block 43. Depot replacement motors (spares).



BLOCK 42

PURPOSE: Block 42 discusses the replenishment of the small electric motor by the User Depot Supply Activity.

PRIMARY PARTICIPANT: User Depot Supply Activity.

SECONDARY PARTICIPANT(S): User Supply Activity, User Depot Maintenance Activity.

RESPONSIBILITIES: The User Depot Supply Activity is responsible for supplying the User's Supply Activity with spare small electric motors. The User Supply Activity is responsible for requisitioning spare motors according to established procedures.

DISCUSSION: The User Depot Supply Activity will provide the User Supply Activity with spare motors as long as they are in stock. Dependent upon supply codes, a failed motor may be discarded at the depot level or it may be overhauled or rebuilt by the depot, or some other activity by depot direction.

When the motor cannot be provided by the User Depot Supply Activity, then the User Depot Maintenance Activity will if possible overhaul or rebuild the motor. Both User and User Depot Supply Activities should be advised of these actions.

Another alternative is for the User Depot Supply Activity to obtain a spare motor from another User Supply Activity. This exchange of User assets will keep systems operational and allow the User Depot Supply Activity to obtain additional assets for all Users, and Depot stock, as necessary.

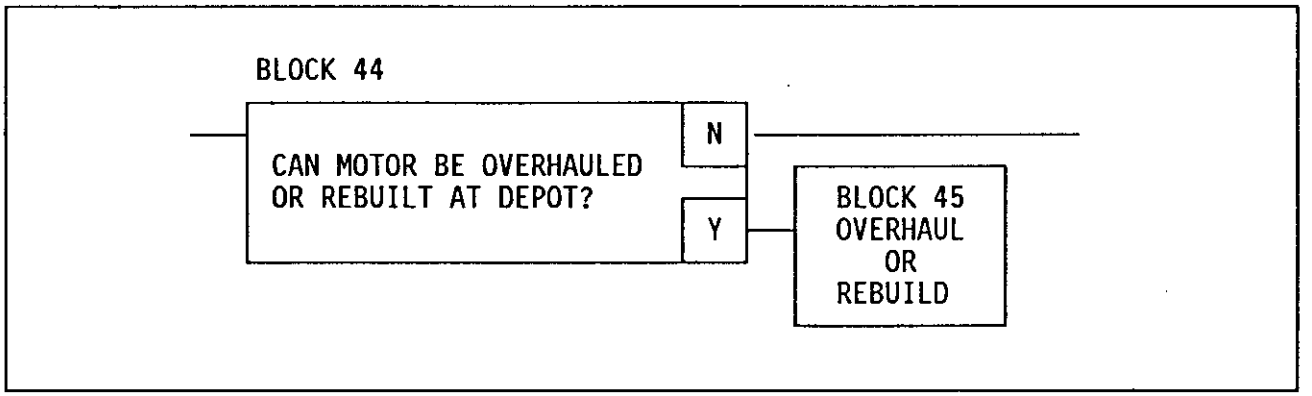
REFERENCES: Departmental guidelines, directives, regulations.

BLOCK 43

PURPOSE: Block 43 is a command block. It requires that a spare small electric motor is on hand, is issued to the User Supply Activity, and is replaced by the User Maintenance Activity. The system is now operational.

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5.3.24 Block 44 and Block 45. Depot overhaul, rebuild actions.



BLOCK 44

PURPOSE: Block 44 concerns overhauling and rebuilding small electric motors at the User Depot Maintenance Activity.

PRIMARY PARTICIPANT: User Depot Maintenance Activity.

SECONDARY PARTICIPANT(S): User Depot Supply Activity.

RESPONSIBILITIES: The User Depot Maintenance Activity is responsible for overhauling and rebuilding small electric motors. These motors are then placed in the motor stocks at the User Depot Supply Activity.

DISCUSSION: If the User Depot Maintenance Activity has a small electric motor overhaul and rebuild capability, then the motor will be inspected for failure mode and will most likely be returned to full service use. Overhaul and rebuild actions will be according to established procedures using correct parts, tools, test equipment, lubricants and production line techniques. Following overhaul and rebuild, and test and acceptance, the motor will be packed and packaged according to specified requirements. It will then be returned to the User Supply Activity.

Depot overhaul and rebuild of motors may not be feasible due to a low turnover in restorable motors. When this is the case, then alternatives must be found to have motors overhauled or rebuilt at other sources.

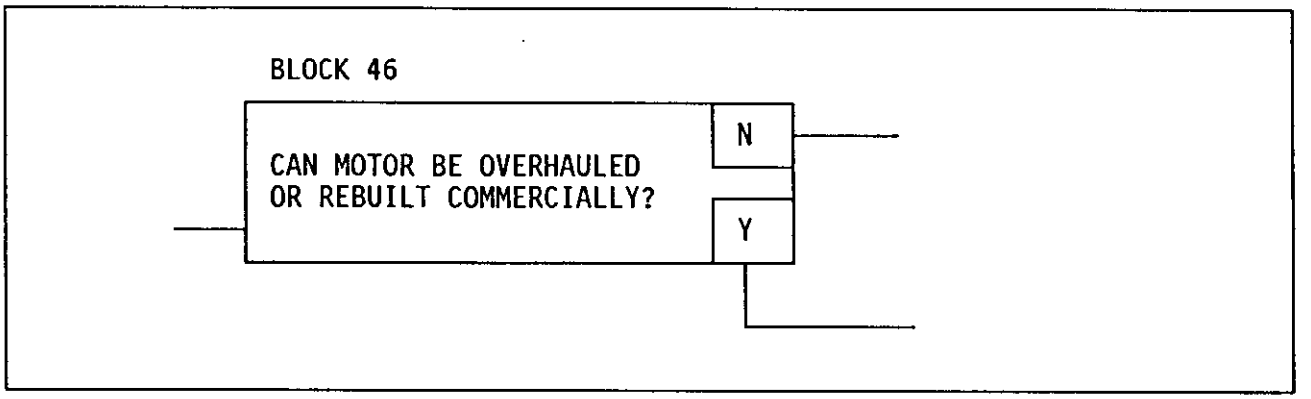
REFERENCES: Departmental guidelines, directives, regulations.

BLOCK 45

PURPOSE: Block 45 is a command block. It assumes that the User Depot Maintenance Activity has a small electric motor overhaul and rebuild capability. The motor will be overhauled or rebuilt, returned to the User Depot Supply Activity, returned to the User Supply Activity, issued to the User Maintenance Activity, and replaced in the new system. The system is now operational.

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5.3.25 Block 46. Commercial repairs.



BLOCK 46

PURPOSE: Block 46 addresses the possibility of having the small electric motor overhauled or rebuilt by a commercial source.

PRIMARY PARTICIPANT: User Depot Supply Activity.

SECONDARY PARTICIPANT(S): User Depot Maintenance Activity.

RESPONSIBILITIES: The User Depot Supply Activity is responsible for maintaining a stock level of small electric motors. The User Maintenance Activity is responsible for Depot overhaul and rebuild decisions.

DISCUSSION: Up to now, in blocks 30 through 45, the new system's small electric motor was placed in operation, maintained, repaired, overhauled, and rebuilt according to established maintenance concepts and procedures. Spare motors for replacement in the new system by the User Maintenance Activity were obtained from the User or Depot Supply Activities. The Depot Maintenance Activity provided overhauled or rebuilt motors for both Depot Supply and User Supply Activity stocks.

Each level of maintenance was authorized to work on specific components. The complexity of work increased with the level of maintenance. Decisions made concerning the level of work and maintenance were the result of establishing a system maintenance concept and provisioning for spare and repair parts. These decisions were based upon:

- Motor design and ability to penetrate to selected components.
- Selection of components to be maintained.
- Tools and test equipment necessary to achieve a satisfactory and lasting repair.
- Personnel and training requirements.
- Reliability, availability, maintainability of the new system.
- Facility requirements and location(s).

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- Cost and time savings in terms of restoring the new system to operational use.

Now, in this block, and for whatever reason, the decision made is that the small electric motor will not be overhauled or rebuilt by the User Depot Maintenance Activity. Two alternatives are open with respect to obtaining spare small electric motors for the new system:

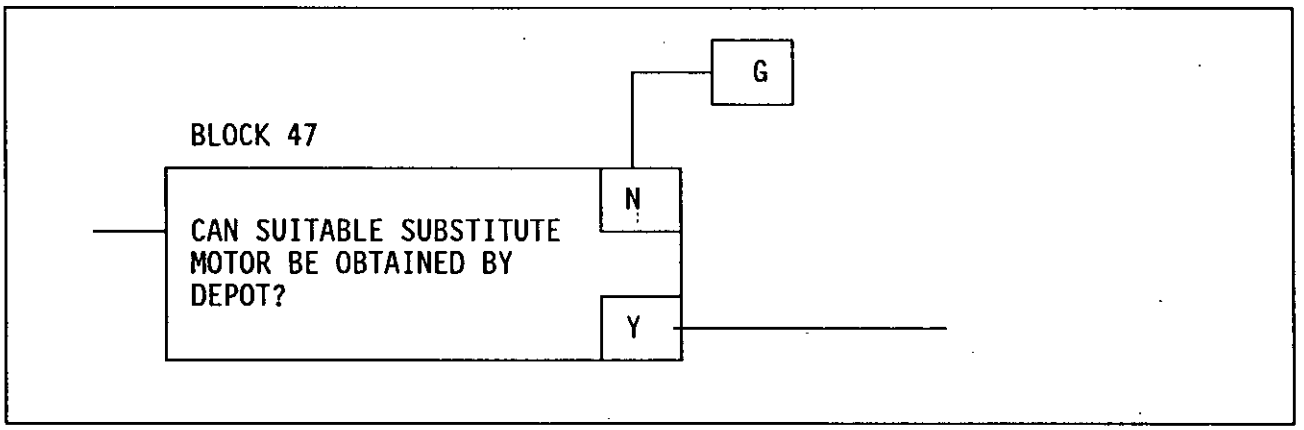
- Use commercial motor repair source(s).
- Identify and use a suitable substitute.

Technical data will be regained so that each alternative can be evaluated and a course of action determined.

REFERENCES: Departmental guidelines, directives, regulations.

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5.3.26 Block 47. Suitable substitute motor.



BLOCK 47

PURPOSE: Block 47 focuses upon the need to find a substitute small electric motor when all sources of supply for the original motor have been, or are about to be exhausted.

PRIMARY PARTICIPANT: Item Manager.

SECONDARY PARTICIPANT(S): User Depot Supply Activity, User Maintenance Activity, User Depot Engineering Activity, Item Manager Engineering Activity.

RESPONSIBILITIES: The Item Manager is responsible for assuring that spare small electric motors are available for new system support requirements. Secondary participants are responsible for assisting the Item Manager as necessary.

DISCUSSION: At this stage of the new system's operational life, the situation exists where continued support of the system is jeopardized by the inability to supply the original small electric motor as required. For whatever reason, there will be no further production of the original motor and its repair parts. Stocks of original small electric motors and repair parts will be depleted in the near future.

The small electric motor Item Manager must know when to address the need to identify and provide a suitable substitute motor for use in the new system. The Item Manager may be a part of a Depot organization, may be collocated with a depot., or may be at another location. Engineering support for the Item Manager may be the responsibility of a Depot or a separate activity.

Faced with the situation to identify a suitable substitute small electric motor, the Item Manager must obtain technical data which identifies the necessary motor characteristics (see block 2 and paragraph 5.3.2). With this data, secondary participants can make the determination as to the availability of a suitable substitute motor.

As the new system becomes older, there is the possibility that a suitable substitute motor may not be available. This happens when the industrial base for an item no longer exists.

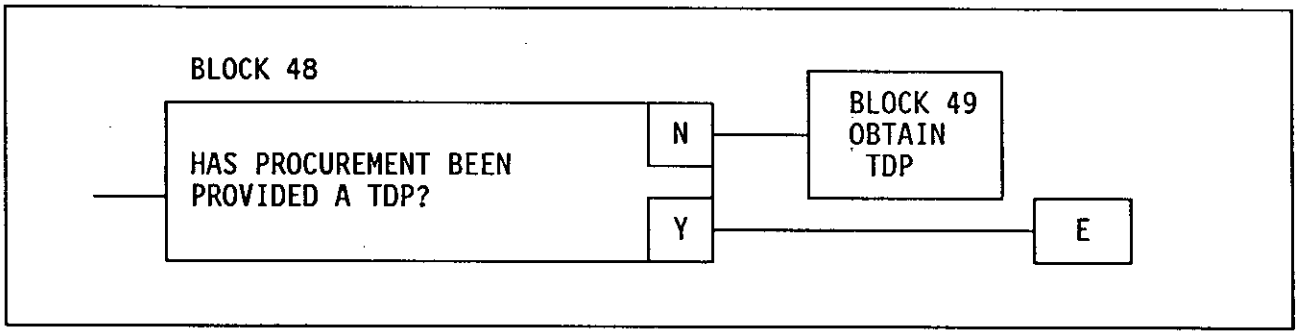
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The remaining blocks of the small electric motor decision network address obtaining a suitable substitute motor or other courses of action.

REFERENCES: Departmental guidelines, directives, regulations.

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5.3.27 Block 48 and Block 49. Substitute motor procurement TDP.



BLOCK 50

PURPOSE: Block 48 addresses forwarding a TDP to the applicable procurement office.

PRIMARY PARTICIPANT: Item Manager.

SECONDARY PARTICIPANT(S): Item Manager, User Depot Maintenance Activity, User Depot Engineering Activity, Item Manager Engineering Activity, Procurement Officer, User Developmental Engineer.

RESPONSIBILITIES: The Item Manager is responsible for forwarding a suitable substitute small electric motor TDP to the applicable government procurement office. The Procurement Officer is responsible for receipt and processing of the TDP according to established schedules. Other secondary participants are responsible to assist the Item Manager in preparing the TDP.

DISCUSSION: Regardless of the Item Manager's organization and location, that manager will be supported in whole or part by the secondary participants or other similar activities. The substitute small electric motor should be prepared with the same amount of care and attention as that of the original motor TDP (see block 6 and paragraph 5.3.5).

When the Item Manager forwards the substitute small electric motor TDP to the Contracting Officer, the bridge between item management and procurement is crossed. Failure to accomplish this task on time can delay the award of a contract for the motor. This delay can, in turn, seriously impact downstream new system operations. Being aware of the schedule, the Procurement Officer must confirm that the TDP was received by procurement and is being properly processed by the procurement staff. Processing the TDP includes checking it for conformance to the format and contents required by directives and regulations.

REFERENCES: Departmental directives concerning preparation, transmission, and receipt of technical data packages.

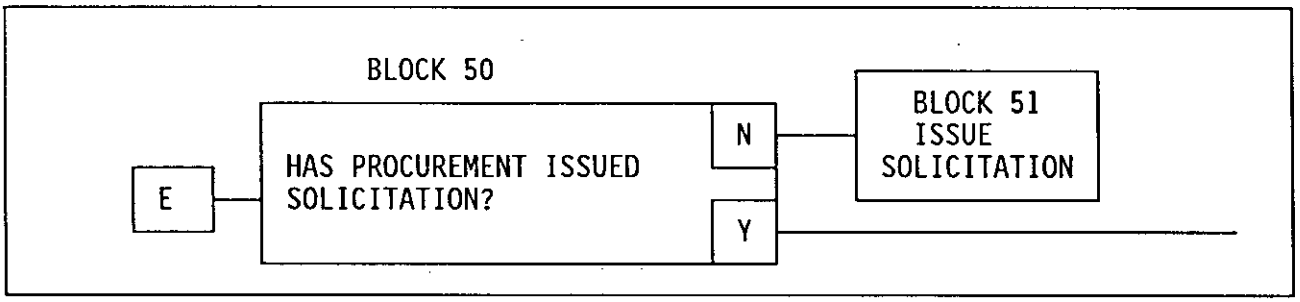
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BLOCK 49

PURPOSE: Block 49 is a command block. It requires that the Procurement Officer obtain the substitute small electric motor TDP from the Item Manager.

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5.3.28 Block 50 and Block 51. Issue substitute motor solicitation.



BLOCK 50

PURPOSE: Block 50 is a tollgate. It assures that the substitute small electric motor solicitation was prepared and issued, or that the issuance action is underway.

PRIMARY PARTICIPANT: Procurement Officer.

SECONDARY PARTICIPANT(S): Item Manager, Depot Engineering and Item Manager Activities.

RESPONSIBILITIES: The Procurement Officer is responsible for preparing and issuing the substitute small electric motor solicitation. All secondary participants will provide full cooperation and assistance.

DISCUSSION: The solicitation for the substitute small electric motor must contain all details necessary to clearly describe what is wanted, when it is wanted, how it is wanted, and why it is wanted. The more data and detail in the solicitation package, the better the responses from small electric motor manufacturers.

Before a solicitation is prepared, the Procurement Officer must decide whether or not to solicit small business firms. The decision is based upon departmental requirements and guidelines for small business procurements.

If the decision is for a small business solicitation, then another decision is required: Whether or not to solicit labor surplus areas and minority small businesses? This decision will be influenced by government guidelines.

Timely preparation of the small electric motor solicitation should result in timely attainment of subsequent steps in acquiring the motor. To this end, the Item Manager and Engineering Activities should assist the Procurement Officer in seeing that the solicitation is prepared on time.

In the event that the Procurement Officer does not need assistance, the Item Manager should track the solicitation to assure it goes out on time. Failure to issue the solicitation on time may adversely impact the new system operations.

REFERENCES: Departmental procurement guidelines, directives, regulations.

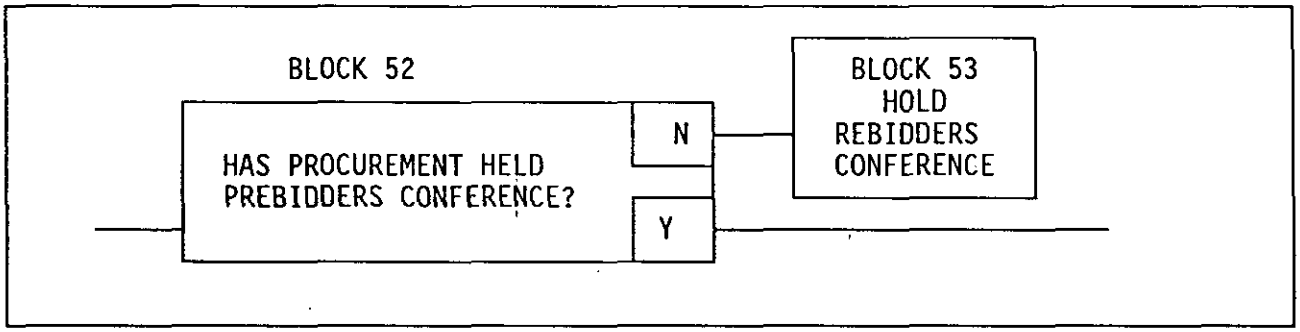
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BLOCK 51

PURPOSE: Block 51 is a command block. It requires that the solicitation be prepared and issued.

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5.3.29 Block 52 and Block 53. Substitute motor prebidders conference.



BLOCK 52

PURPOSE: Block 52 identifies the use of a prebidders conference for the new system's substitute small electric motor.

PRIMARY PARTICIPANT: Procurement Officer.

SECONDARY PARTICIPANT(S): Item Manager, User Depot Engineering Activity, Item Manager Engineering Activity, Potential substitute small electric motor suppliers.

RESPONSIBILITIES: The Procurement Officer is responsible for assuring that potential small electric motor manufacturers have the opportunity to jointly attend a conference concerning the solicitation. Engineering activity members such as the Motor Designer and System Design Engineer should attend this conference to supply technical information. Potential suppliers should attend the conference to improve their understanding of the solicitation.

DISCUSSION: A prebidders conference is not always required. But, by scheduling and holding one the Procurement Officer and secondary participants can easily see who the potential suppliers are, and what their understanding of the solicitation is.

During a prebidders conference, potential suppliers have the opportunity to ask questions and obtain responses. Invariably, a transfusion of information occurs, and the suppliers gain a better understanding of the solicitation. At times, a solicitation is amended due to the need to insert clarifying statements within.

Some suppliers lose interest, while others decide to compete for the contract award. Discussions cover a wide range of topics, including those for:

- Availability of contractor and government specifications and standards.
- Test data and reports.
- Special test requirements.
- Use of end article (multiple use?).
- Maintenance requirements.
- Preaward surveys.
- Quality assurance.
- Substitute items.
- Former government surplus items.
- Schedules.
- Payments, progress payments.

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- Proof tests.
- First article qualification tests.
- Qualification tests.
- Schedules.
- Prime contractor and government contract administration.

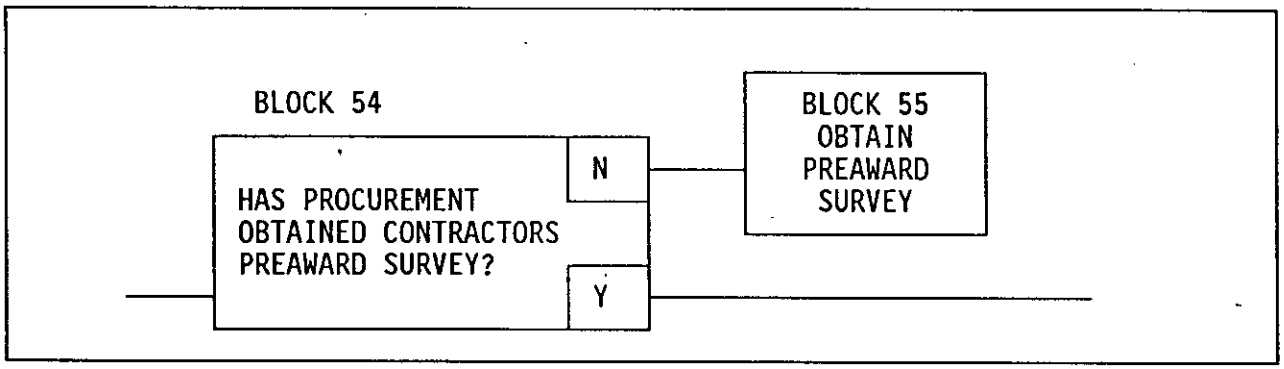
REFERENCES: Departmental procurement guidelines, directive, regulations.

BLOCK 53

PURPOSE: Block 53 is a command block. It requires that a Prebidders conference be held when necessary.

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5.3.30 Block 54 and Block 55. Substitute motor preaward survey.



BLOCK 54

PURPOSE: Block 54 concerns the selection of a capable and competent supplier from those who submitted bids for the system's substitute small electric motor.

PRIMARY PARTICIPANT: Procurement Officer.

SECONDARY PARTICIPANT(S): Item Manager, User Depot and Item Manager Engineering Activities.

RESPONSIBILITIES: The Procurement Officer is responsible for conducting a preaward survey. Secondary participants join ins the survey, either by directive or by choice.

DISCUSSION: A preaward survey is conducted for the purpose of aiding the Procurement Officer in making the decision as to which bidder will be finally selected for any given contract award. By carefully reviewing bidders' inputs and making use of all available information, the Procurement Officer can usually narrow the number of responsive and responsible bidders down to the single one most likely to be able to complete the contract satisfactorily. When confirmation is desired or required, the Procurement Officer initiates the preaward survey process.

Some preaward surveys can be accomplished by a "desk audit." This action usually covers those suppliers who have constantly produced satisfactory end articles for use by their customers. A check on plant loading may be all that is required for confirmation of a decision to award a contract to that supplier.

On the other end of the spectrum is the need for a complete preaward survey. For whatever reason, the Procurement Officer wants an in-plant check of the potential contract winner. Items to be checked include plant management, plant administration, financial stability and history, present debt, plant loading, product history, product quality assurance, delivery schedules, adequate and properly strained work force, and current production of the same or similar items.

The procurement Officer designates experienced persons to check each area of interest. Other persons may request to participate in the survey. Each input rates the potential supplier as to being capable and responsive to contract requirements. Finally, each preaward designated participant makes a recommendation, fully justified, as to whether or not the potential supplier should be awarded the

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contract. This information is used by the Procurement Officer to make the final yes or no decision concerning the contract award.

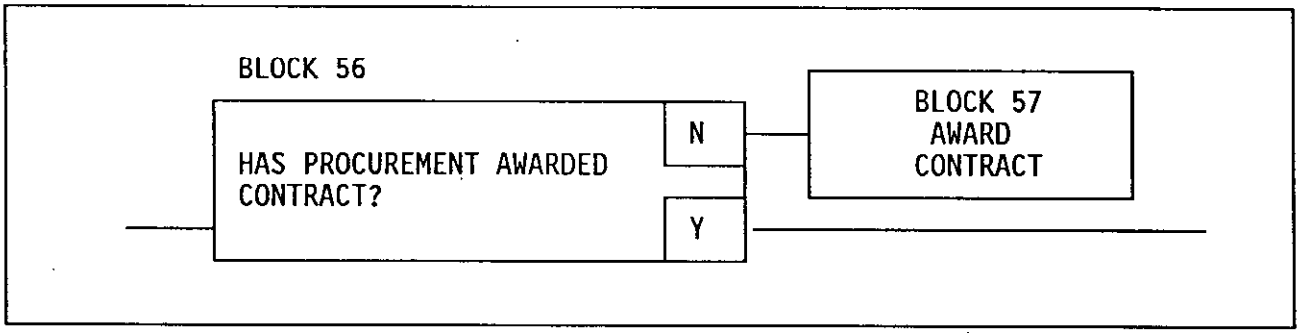
REFERENCES: Departmental preaward survey guidelines, directives, regulations.

BLOCK 55

PURPOSE: Block 55 is a command block. It requires that a preaward survey be used by procurement staff.

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5.3.31 Block 56 and Block 57. Substitute motor contract award.



BLOCK 56

PURPOSE: The purpose of block 56 is to highlight the preparation and award of the substitute small electric motor contract.

PRIMARY PARTICIPANT: Procurement Officer.

SECONDARY PARTICIPANT(S): User Depot and Item Manager Engineering Activities, Motor Supplier.

RESPONSIBILITIES: The Procurement Officer is responsible for developing a contract to be offered to the Motor Supplier. Engineering Activities provide technical input to the Procurement Officer. The Motor Supplier is responsible for understanding the terms and conditions of the contract, and for accepting or otherwise coming to an agreement with the Procurement Officer.

DISCUSSION: A contract is the visible result of an offer, acceptance, and mutual consideration. Put another way, a contract represents a combination of (1) form, (2) compensation arrangement, and (3) end purpose; e.g., a purchase order (form) with a fixed price (compensation arrangement) for supplies (end purpose).

The award of the contract is the end result of a lot of preparatory activity by the primary and secondary participants. For the Procurement Officer, blocks 48-54 set the stage for contract award. For the winning motor supplier, the activity begins when the solicitation is received.

Both participants must work to their internal schedules: Procurement Officer - motor delivery schedule; motor supplier - motor delivery schedule. Cost and pricing data is developed by both participants. The price is reviewed by the Procurement Officer, and a price analysis may be performed to determine the fairness of the price. Finally, when all details are agreeable to both participants, the contract is completed and awarded to the motor supplier.

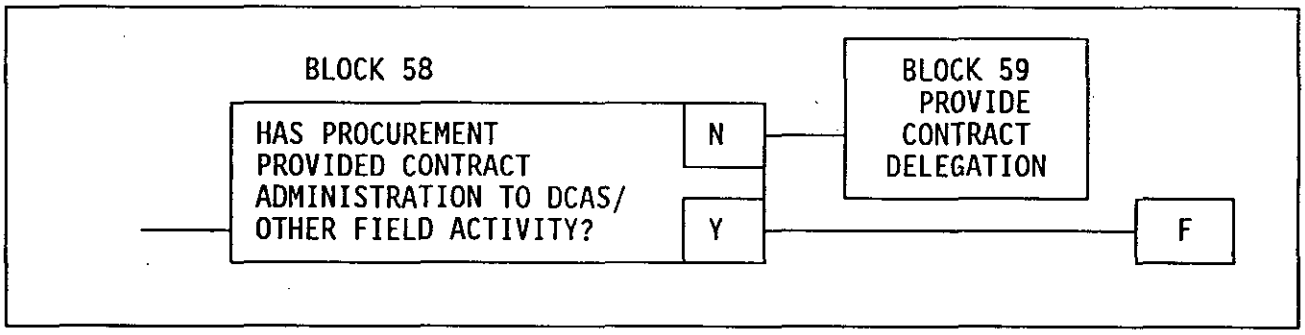
REFERENCES: Government and Motor Supplier's guidelines, directives, regulations.

BLOCK 57

PURPOSE: Block 57 is a command block. It serves as a check point to assure a contract is awarded within schedule requirements.

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5.3.32 Block 58 and Block 59. Substitute motor contract administration.



BLOCK 58

PURPOSE: The purpose of block 58 is to provide a brief on government contract administration.

PRIMARY PARTICIPANT: Procurement Officer.

SECONDARY PARTICIPANT(S): DCAS, other government contract administration activity.

RESPONSIBILITIES: The Procurement Officer is responsible for delegating administration of the substitute small electric motor contract. DCAS is responsible for contract administration at the motor supplier's facility and external supplier levels of activity. Government contract administration focuses principally upon the motor supplier, and at times on other suppliers.

DISCUSSION: The contract administration function is the focus of all activity between contracting parties. Some of the many facets of contract administration are:

- Contract amendments or changes.
- Subcontracts and purchase orders.
- Cost and price analysis.
- Property control.
- Production control.
- Quality assurance, quality control.
- Subcontractor proposal reviews.
- Subcontract negotiations.
- Labor law compliance.
- Correspondence and reports.
- Allowable costs.
- Safety.
- Security.

At times the Procurement Officer may perform contract administration functions with an internal staff, or by delegation to DCAS or other contract administration activities.

Government contract administration is usually limited to those contractors having government contracts. But, the government retains the right to check any or all

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steps of manufacturing of its material at subcontractor plants with or without priority of contract. Arrangements for government inspection are made by both the government-prime contractor route and government-subcontractor route. In the latter case, the government procurement office usually delegates the inspection of the subcontractor to DCAS, or some other government contract administration function. DCAS, for example, with then notify the subcontractor of the need for government inspection. The prime contractor also notifies the subcontractor of the government intent. Subsequent to these notifications, the government then performs its inspection as necessary.

Separate inspections of subcontractors by both the prime contractor and the government usually assure that acceptable products are delivered to the government. Product deficiencies that occur may be traced to:

- Poor TDP's.
- Insufficient inspections and tests.
- Unskilled inspectors.
- Inadequate production controls.
- Unskilled work force.
- Awards to incompetent manufacturer.
- Poor preaward surveys.

Proper feedback of contract and contractor deficiencies for review and correction by contract administration activities should reduce both future problems, occurrences, and receipt of deficient products for government use.

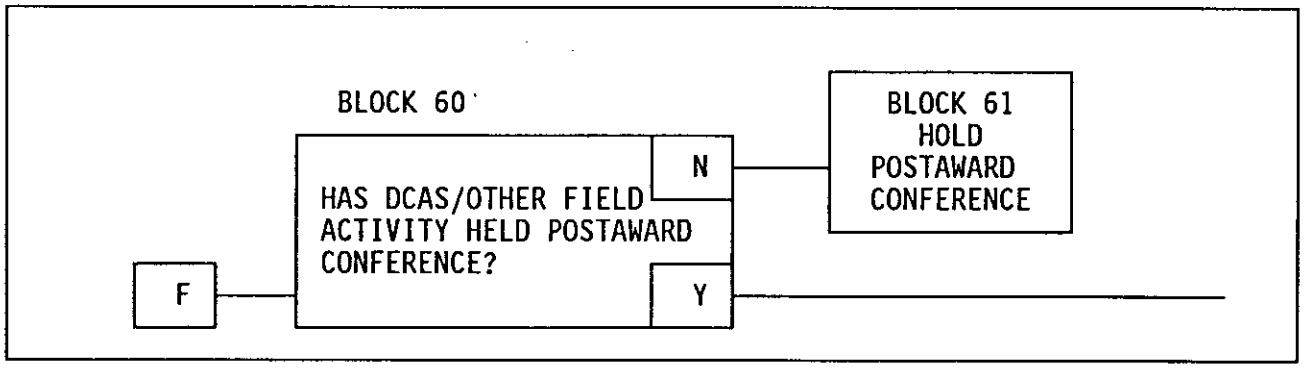
REFERENCES: Departmental guidelines, directives, regulations; contract requirements.

BLOCK 59

PURPOSE: Block 59 is a command block. It requires that contract administration delegations be made for field surveillance of the substitute small electric motor supplier's contract performance.

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5.3.33 Block 60 and Block 61. Substitute motor postaward conference.



BLOCK 60

PURPOSE: The purpose of block 60 is to again identify the substitute small electric motor postaward conference function and provide some information of its activity.

PRIMARY PARTICIPANT: Procurement Officer.

SECONDARY PARTICIPANT(S): Item Manager, Depot and Item Manager Engineering Activities, DCAS, other contract administration activity, Motor Supplier.

RESPONSIBILITIES: The Procurement Officer is responsible for making arrangements and holding the substitute small electric motor contract postaward conference. DCAS or another government contract administration activity may attend by request or by choice. Other secondary participants are parties to the conference.

DISCUSSION: During the postaward conference all participants discuss the substitute small electric motor contract, its requirements to include schedules, and the functions and responsibilities of each party. Questions are raised and answered, and procedural problems are resolved. Surveillance actions by both the Procurement Officer and DCAS, or other designated contract administration activity, are explained as necessary to the Motor Supplier.

The results of this conference should leave no question unanswered concerning the function, conduct, and mutual and individual responsibilities for successful contract performance.

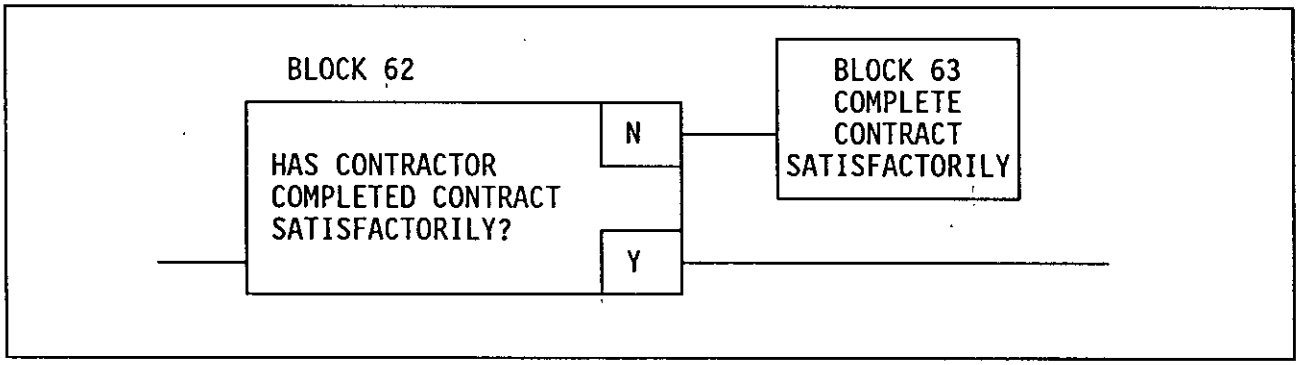
REFERENCES: Departmental guidelines, directives, regulations; contract requirements: contract delegation.

BLOCK 61

PURPOSE: Block 61 is a command block. It requires that a postaward conference be held, preferably before the substitute small electric motor supplier commences work.

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5.3.34 Block 62 and Block 63. Substitute motor contract completion.



BLOCK 62

PURPOSE: The purpose of block 62 is to highlight the satisfactory completion of the substitute small electric motor contract.

PRIMARY PARTICIPANT: Motor Supplier.

SECONDARY PARTICIPANT(S): Procurement Officer, DCAS, other contract administration activity.

RESPONSIBILITIES: The Motor Supplier is responsible for the satisfactory completion of the substitute small electric motor contract. Secondary participants are responsible for tracking the Motor Supplier's progress through contract completion.

DISCUSSION: The satisfactory completion of a contract involves more than government acceptance of contract end articles. Depending upon contract requirements, the government will almost always require other contract deliverables such as drawings, TDP's, reports, inspection data, technical manuals, and financial and progress reports.

Government in-process hardware inspections at its contractor's and subcontractor's facilities are some of the most important steps in assuring satisfactory contract completion. These inspections should provide data that confirm satisfactory performance by the contractor or subcontractor. All government data should be used to determine whether or not subcontractor is performing satisfactorily.

Each contract deliverable has a time and place for delivery and customer acceptance. Regardless of when and where scheduled deliveries and acceptances occur, a contract will not be satisfactorily completed unless all items are delivered on time and accepted by the government at that time.

Subcontractor milestone delays may be costly to the Motor Supplier, if due to poor performance, in terms of penalty payments imposed by the contract. And, each unsatisfactorily completed contract milestone becomes, or should be a statistic for the government's contractor performance file. These unsatisfactory actions may require contractual adjustment before a contract can be considered as being completed satisfactorily.

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Finally, all contract participants should work together as much as possible to assure the satisfactory completion of a contract. Early identification and correction of problems almost always will prevent or reduce downstream contractual delays which adversely impact satisfactory contract completion.

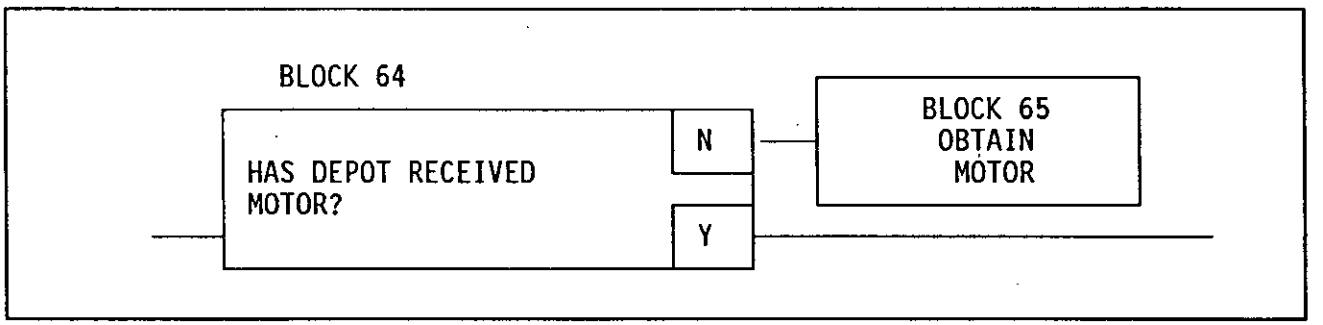
REFERENCES: Contract requirements; government's and motor supplier' guidelines, directives, regulations; DCAS and other field activity guidelines, directives, regulations.

BLOCK 63

PURPOSE: Block 63 is a command block. It requires all contract requirements to be met for the small electric motor contract to be considered satisfactorily completed.

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5.3.35 Block 64 and Block 65. Substitute motor receipt.



BLOCK 64

PURPOSE: The purpose of block 64 is to direct attention to the receipt of the substitute small electric motor at the User Depot.

PRIMARY PARTICIPANT: User Depot Supply Activity.

RESPONSIBILITIES: The User Depot Supply Activity is responsible for the receipt and accountability of the substitute small electric motor. The Motor Supplier must ship the substitute motor as directed by contract.

DISCUSSION: Shipping the new system's substitute small electric motor from the Motor Supplier is just the beginning of the motor receipt process.

The Motor Supplier must ship the motor according to contract requirements. Usually a contract will stipulate levels of packaging and packing, mode of transportation, destination, and required receipt date. Part of the shipping process may include notifying the User Depot Supply Activity of the shipment date and the name of the cargo carrier.

Upon receipt of notification of motor shipment, the User Depot Supply Activity prepares for the motor's arrival. Receipt procedures should include inspection of the motor shipment for count and condition of packaging and packing. If there is a shipment problem related to motor damage, it has to be resolved by the carrier, motor supplier, and User Depot.

Following receipt of the motor at the User Depot, there may be the requirement for bench tests before shipment to the User Supply Activity. Also, concurrent inspection of accompanying technical data, such as technical manuals and handbooks, should be accomplished. All discrepancies should be resolved by the Item Manager and Motor Supplier.

Finally, the User Depot and User Supply Activities concerned with the small electric motor must be notified of its arrival, status, and location.

REFERENCES: Contract requirements; Departmental and Motor Supplier guidelines, directives, regulations.

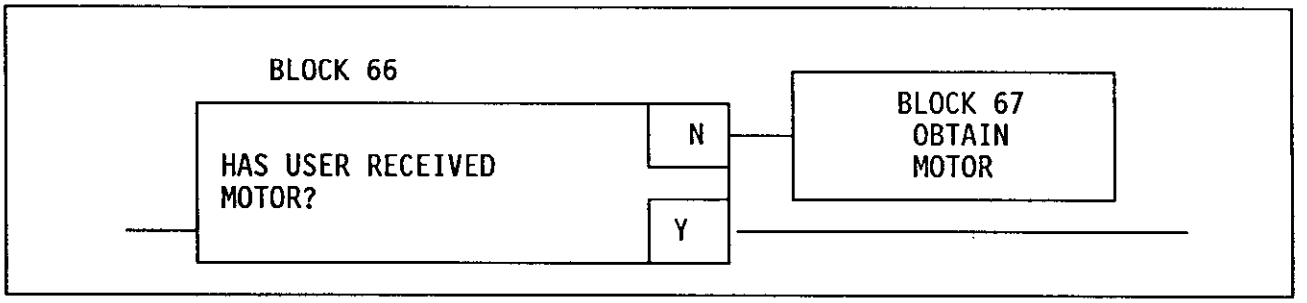
BLOCK 65

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PURPOSE: Block 65 is a command block. It requires the Item Manager to take aggressive action to see that the motor is on hand when required.

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5.3.36 Block 66 and Block 67. User receipt of substitute motor.



BLOCK 66

PURPOSE: Block 66 highlights the movement of the substitute small electric motor from the User Depot Supply Activity to the User Supply Activity.

PRIMARY PARTICIPANT: User Supply Activity.

SECONDARY PARTICIPANT(S): User Depot Supply Activity.

RESPONSIBILITIES: The User Supply Activity is responsible for receiving and accounting for the substitute small electric motor. The User Depot Supply Activity must notify the User Supply Activity of the motor shipment information; i.e., date, carrier, arrival time, etc.

DISCUSSION: Shipping the substitute small electric motor from the User Depot Supply Activity is just the beginning of the motor receipt process.

The User Depot Supply Activity must ship the motor according to their established requirements. Subjects of consideration include: mode of transportation such as air, rail, truck; insurance; schedule; carrier; destination; and notification to recipient. By paying close attention to shipping requirements, the User Depot Supply Activity can substantially reduce the chances of shipment problems occurring.

Following notification of motor shipment, the User's Supply Activity prepares for the motor's arrival. Receipt procedures should include inspection of the substitute motor shipment for count and condition, and bench tests for satisfactory operation. If there is a problem then it has to be worked out between the carrier, User Depot, and User Supply Activities.

After satisfactory receipt the substitute motor may be stored until required for test and installation. All User activities concerned with the motor must be notified of its arrival, condition, and location.

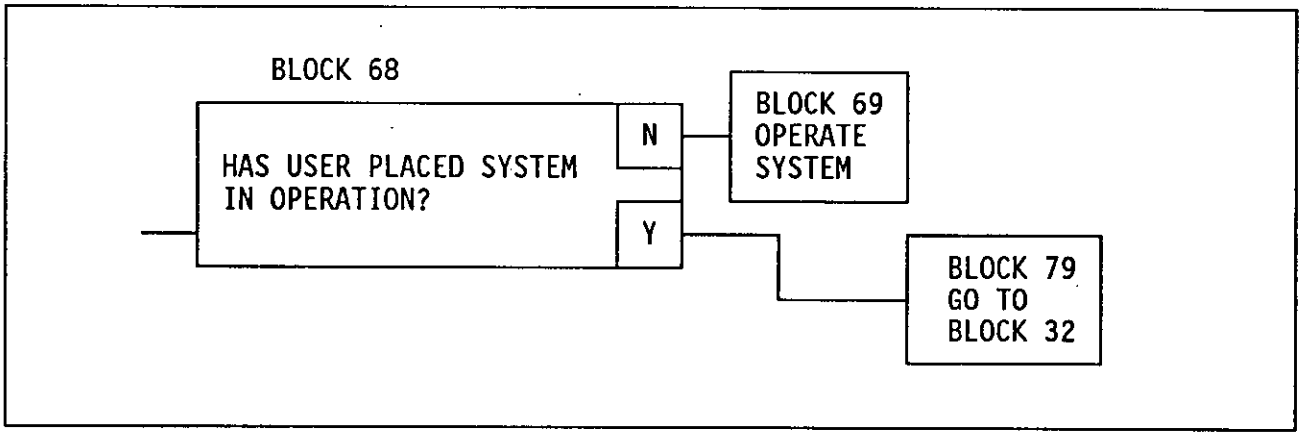
REFERENCES: Departmental guidelines directives, regulations.

BLOCK 67

PURPOSE: Block 67 is a command block. It requires the User to take aggressive action to see that the motor is on hand when required.

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5.3.37 Block 68, Block 69, and Block 70. Small motor replacement and new system operation.



BLOCK 68

PURPOSE: Block 68 concerns restoring the new system to full operational use as a result of the replacement of the failed original small electric motor with a substitute motor.

PRIMARY PARTICIPANT: User.

SECONDARY PARTICIPANT(S): User Maintenance and User Supply Activities.

RESPONSIBILITIES: The User Maintenance Activity obtains the substitute small electric motor from the User Supply Activity. After motor replacement, the User operates the new system.

DISCUSSION: After being notified of the receipt of the substitute electric motor by User Supply, User Maintenance will test the motor, replace it in the new system, and accomplish new-system check-out procedures. Following a satisfactory new system check-out, the User places the system in full operation.

REFERENCES: Departmental guidelines, directives, regulations.

BLOCK 69

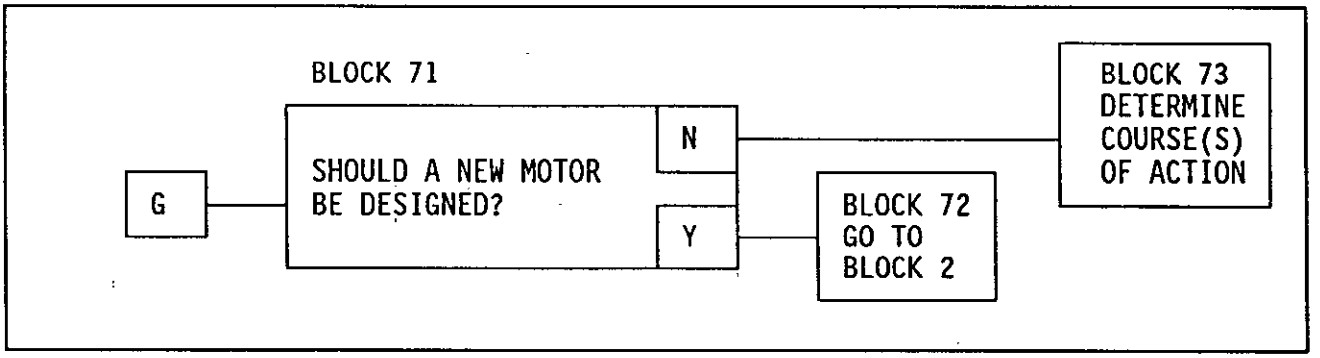
PURPOSE: Block 69 is a command block. It requires that all User activities complete all necessary actions to attain system operational capability, and to sustain system operations from then on.

BLOCK 70

PURPOSE: The purpose of block 70 is to transfer attention to the fact that the restored, fully operational new system continue its operational role at block 32. The current new system will continue its useful life by cycling through blocks 32 through 70.

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5.3.38 Block 71, Block 72, Block 73. New motor design.



BLOCK 71

PURPOSE: Block 71 highlights some activities necessary to decide whether or not a new motor design is feasible for use in the new system when a suitable substitute motor is not available (see block 47).

PRIMARY PARTICIPANT: Item Manager.

SECONDARY PARTICIPANT(S): User Depot Supply Activity, User Maintenance Activity, User Depot Engineering Activity, Item Manager Engineering Activity, Developmental Engineer (User), Motor Designer.

RESPONSIBILITIES: The Item Manager is responsible for making the decision concerning the feasibility of designing a new motor. Secondary participants will assist the Item Manager as required.

DISCUSSION: Principal events leading to block 71 are shown in blocks 46 and 47. Activities for these blocks lead to the conclusion that there is now no capability to support the original new system with spare original motors or suitable substitutes.

A very important meeting occurs in block 71. At this meeting the Item Manager and secondary participants discuss the feasibility of designing a new motor to keep the system operational. Topics which must be discussed include: system phase-out date, industrial base from which to obtain a new motor, economics of maintaining the original new system, feasibility of product improvement program for old system, and more.

REFERENCES: Departmental guidelines, directives, regulations.

BLOCK 72

PURPOSE: Block 72 is a command block. It moves the new system requirement for a new small electric motor design back to the System Design Engineer. A new motor design will be obtained by following the network beginning at block 2 (see paragraph 5.3.2.

BLOCK 73

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PURPOSE: Block 73 is a command block that is a real crunch point. It requires that the decision not to design a new motor be reevaluated. If the decision holds firm then the possible courses of action are: (1) phase out system, (2) redesign old system, and (3) design new system. Item (3) would start at block 1 with a new design required. Item (2) would start at block 2. Item (1) implies no further need for system -- declare it nonsupportable and obsolete. At this time, the new system would become government surplus if there is no further use for the system by authorized agencies. Also, depending upon components of the system, it may have to be demilitarized, reduced to scrap, or destroyed.

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6. NOTES

6.1 Intended Use. By following the small electric motor decision network shown within this handbook the reader should be able to gain or expand an appreciation of the small electric motor life cycle, and thus help prevent or correct immediate problems associated with field failures of these motors.

6.2 Subject term (key word) listing.

Contract(s)
Electric motor characteristics
Contract award
Motors, electric
Motors, electric, small
Postaward conference
Preaward survey
Technical data
Technical data package

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Custodians:
Army - ME
Air Force - 99

Review activities:
Army - AV
Navy - SH
Air Force - 84
DLA - GS

User activities:
Air Force - 80, 71, 70

Agent:
Army - ME

Preparing Activity:
OUSD-SD
(Project 6105-0150)

Agent:
Army - ME

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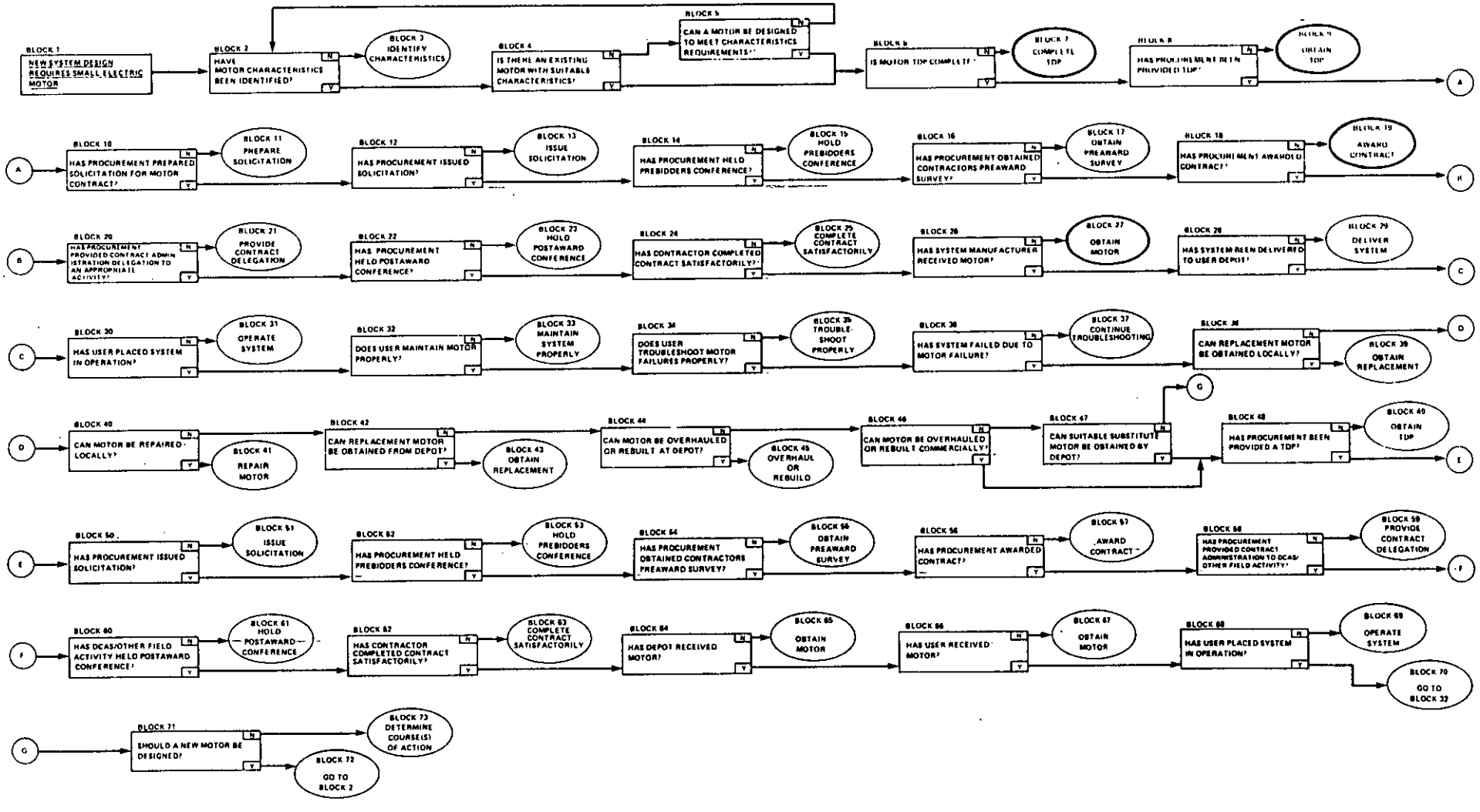


FIGURE 1. Small electric motor decision network.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-HDBK-117	2. DOCUMENT DATE (YYMMDD) 910401

3. DOCUMENT TITLE
Motors, Electric, Small Acquisition, Operations and Support for Primer

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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